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Hybrid machine learning and MCDM framework for consumer preference extraction and decision support in dynamic markets

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ABSTRACT

With the rapid development of e-commerce and digital consumption, online reviews have become an important channel for consumers to express their opinions and for businesses to understand market dynamics. However, the surge in review volume—resulting in information overload and a large amount of irrelevant content—has severely hindered the accurate identification of genuine consumer preferences, directly affecting the accuracy of product design, marketing, and resource allocation decisions. To address this challenge, this paper proposes an innovative hybrid framework that integrates information entropy, a binary classification model, PCA combined with K-means clustering, the BERT-wwm-ext sentiment analysis model, and multi-criteria decision-making (MCDM) methods, aiming to enhance the accuracy of preference analysis and the reliability of decision-making in the context of digital consumption. The framework tackles three key challenges: bias in traditional evaluations of perceived useful information, incompleteness in preference feature extraction, and inaccuracies in preference weight calculation. A comprehensive analysis of over 70,000 online customer reviews sourced from platforms such as the Apple App Store and JD.com validates the framework, showing that it outperforms existing models in predicting perceived usefulness, uncovering hidden product attributes, and refining feature weight calculations. This study not only provides robust data support for enterprises in product optimization and targeted marketing, but also offers decision makers a scientifically grounded framework for product management and efficient resource allocation that better aligns with consumer needs.

1. Introduction

The development of big data analytics and related technologies has made it feasible to extract the performance, significance, and impact on consumer satisfaction of product attributes from online reviews (Jiang et al., 2025). The vast amount of dynamic feedback on these products provides a critical basis for companies to accurately understand consumer needs and to precisely develop product design and marketing strategies. Consequently, consumer-oriented, accurate product design and marketing are viewed as a panacea for competing in highly competitive markets (Jiang et al., 2023). However, the exponential increase in reviews comes with information overload and a myriad of invalid data (Borghi & Ratcharak, 2025), which severely interferes with companies' ability to accurately extract genuine consumer preferences, thereby making it more challenging to resolve the uncertainty between product development and sales (Alshawawreh et al., 2024). This uncertainty refers to whether a product's features and functions align with consumers' preference requirements (Zhang et al., 2024). For example, Ford received strong criticism on automotive forums for emphasizing an increase in bolt size during vehicle development in its advertisements and claiming to be the largest among all manufacturers. This criticism even prompted competitor Toyota to respond by stating that Toyota focuses more on engines rather than wasting time on bolts (Caldieraro & Cunha, 2022). As this anecdote illustrates, resolving the uncertainty between product development and sales essentially requires addressing whether product functions are consistent with consumer preferences.

While accurately extracting the truly valuable information that consumers care about from the overwhelming array of data holds significant theoretical and practical significance for enhancing corporate decision support, optimizing product design, and refining marketing strategies (Song et al., 2025), there remain three key challenges in accurately conducting preference analysis on online reviews to address the uncertainty between production and sales.

First, although techniques such as machine learning have made big

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0160-791X/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

data analysis feasible, the usefulness of online review information still largely depends on traditional voting methods. However, research has shown that this mechanism suffers from the following biases: 1) Early bird bias (Fresneda & Gefen, 2019), where earlier reviews are more likely to receive exposure and votes, potentially causing higher-quality later reviews to be overlooked; 2) Imbalanced voting bias (Fresneda & Gefen, 2020), meaning that user behavior may lead to certain types of reviews receiving concentrated votes; and 3) Winner cycle bias (Wang et al., 2022), where high-vote reviews tend to attract even more votes in a self-reinforcing cycle, resulting in other useful reviews receiving insufficient exposure. Therefore, accurately extracting consumer perceived useful information (CPUI) from online reviews amidst today's information overload remains one of the critical challenges that must be addressed.

Second, consumer preference analysis is typically based on weighted conjoint analysis of product features. This means that the accurate extraction of product features is crucial for obtaining reliable results in consumer preference analysis (Choi & Kim, 2025). However, conventional methods often fail to uncover hidden attributes due to interference from invalid data (Dong, Yao, et al., 2025), which adversely affects the accuracy of the measured preferences in conjoint analysis (Zhao et al., 2025). Therefore, in the context of big data, comprehensively mining product features that align more closely with the consumer perspective remains one of the key challenges that must be addressed.

Finally, the ranking of feature importance (i.e., preference weight calculation) generally refers to the utility function that reflects an individual's preferences for different alternatives, where the ranking among available options implies the maximization of this utility function (Halder, 2025). However, all information contains a degree of redundancy, which is related to the occurrence probability or uncertainty of each symbol (such as numbers, letters, or words) within the data (Zaidan et al., 2025). Consequently, factors like errors in computing the frequency of non-informative features in traditional weighting methods can impair the accuracy of the calculated weights (Gaurav et al., 2025). Therefore, deriving weight calculations from online review texts that more accurately reflect the consumer perspective remains one of the critical challenges that must be addressed.

Although addressing the uncertainty between production and sales in the era of big data presents key challenges, overcoming them not only enriches the theoretical support for production optimization but also enables more precise decision-making in demand response. First, developing an analytical model that accurately extracts consumer-perceived useful information improves the identification of genuine consumer preferences, thereby offering companies more reliable decision support for product design and marketing strategy formulation. Meanwhile, comprehensively mining product features that align with consumer perspectives helps firms promptly capture shifts in market demand, drive innovation, and reduce decision-making risks. Moreover, weighting methods that consider perceived usefulness objectively reflect consumers' true preferences for various product attributes, thereby optimizing resource allocation, enhancing customer satisfaction and brand loyalty, and ultimately strengthening a company's core competitiveness in highly competitive market environments.

Given the theoretical and practical value that solving the above challenges may bring, this article will address the relevant challenges through the following contributions.

Firstly, to mitigate the three bias issues inherent in traditional voting methods for perceiving useful information, we introduce a prediction model that combines machine learning binary classification with information entropy principles. This model performs an unordered full-text evaluation, ensuring that every review—including those without votes—is considered, and it significantly improves the accuracy and comprehensiveness of classifying textual information.

Secondly, to address the insufficient extraction of product features, particularly the oversight of emerging attributes and implicit consumer demands, we propose a hidden feature mining method based on Consumer-Perceived Useful Information (CPUI). This method employs dynamic perception modeling driven by information entropy and integrates a machine learning algorithm to capture real-time feedback on product iterations, updates, and emerging usage scenarios, thereby outperforming common approaches in both accuracy and the comprehensiveness of the analytical dimensions.

Finally, to overcome the interference of invalid information in traditional weighting methods, we present a novel feature weight computation approach. By deeply integrating the BWM-TOPSIS hybrid decision-making framework with NLP-based noise filtering and optimizing weight allocation through information entropy, our method significantly improves weight stability and ranking accuracy—ensuring that computed weights closely reflect consumers' perceived useful preference demands, and thus providing a novel theoretical foundation and practical methodological support for precise resource allocation, product design, and marketing strategies.

2. Literature review

2.1. Online reviews and big data analytics technology

In the e-commerce environment dominated by online platforms, online reviews have emerged as a vital method for the dissemination of product information (Mladenovic et al., 2024), significantly influencing consumer purchasing behaviors (Chaudhuri et al., 2024). Consumers are able to share their post-use experiences or service feedback through online reviews, which are perceived as "more credible product evaluations published on company or third-party websites"(Roy et al., 2024). Therefore, potential buyers often rely on online reviews to uncover unseen product attributes, aiding their purchasing decisions. Moreover, online reviews offer an extensive amount of information on consumer attitudes towards product features (Zhu et al., 2024), becoming an increasingly critical source of insight for brands and merchants to discern customer preferences and cultivate a brand image (An et al., 2025). This assists companies in understanding consumer perspectives, optimizing marketing-related decision-making processes (Xu & Hu, 2024). The vast amount of data from online reviews addresses the deficiencies of traditional market survey questionnaires, such as insufficient data and a lack of genuine product usage experiences (Chen et al., 2021). Consequently, in recent years, a significant number of researchers have focused on digital transformation studies of marketing activities based on online reviews (Luo et al., 2024).

The large amount of online review information data for a product is referred to as "a massive variety of data types produced at high velocity from numerous sources" (Jap et al., 2022), These big data are primarily characterized by their velocity of generation, diversity, volume, and the structured and unstructured nature of data sources, all growing at an exponential rate (Zablith, 2024). Currently, the utilization of big data analytics on extensive review information is gradually becoming a core marketing tool for many companies, carving out new vistas for the scientific work in marketing (Leon et al., 2024). For instance, Shaik et al. (2024) argue that big data transcends traditional empirical methods, enabling marketing to operate in a more powerful and innovative manner. Feng et al. (2024) posited that big data marketing approaches represent the future of the retail industry.

2.2. The studies of information usefulness and consumer preferences

Multi-criteria decision-making (MCDM) has been widely recognized and applied for its robustness and effectiveness in conjoint analysis and other preference calculation methods, enabling more precise capture and evaluation of the trade-offs and uncertainties among various features or indicators. For example, Abdel-Basset et al. (2024) proposed an integrated MCDM framework that combines AHP and TOPSIS; by accurately computing feature weights, the framework addresses uncertainty and ambiguity in the decision-making process and effectively

evaluates the impact of Industry 5.0 technologies. Petrudi and Arabi (2025) introduced an MCDM research framework based on the Group Grey-Best-Worst Method (GG-BWM) to develop strategies for circular supply chains (CSC) in providing sustainable products to users, and they validated its effectiveness. Liu et al. (2024) constructed an MCDM model to capture tourists' expectations of different tourism product attributes, offering a novel approach for tourism companies to design personalized tourism products that better meet customer needs.

In recent years, machine learning methods have also been widely applied in feature importance assessment and preference prediction studies. For instance, Chang (2024) employed models such as XGBoost and KMV to construct a credit risk evaluation framework, effectively identifying the credit risk characteristics of technological SMEs in China through precise feature selection. Moreover, Chang et al. (2024) utilized machine learning alongside ARIMA time series models to forecast tourists' preferences for attractions, demonstrating that machine learning approaches can accurately extract consumer preference characteristics in dynamic environments. These studies further corroborate the effectiveness and promising prospects of integrating machine learning with MCDM for preference research, as adopted in this paper.

It is noteworthy that the accuracy of product feature preference weighting is directly related to the accuracy of feature extraction (Zhang & Niu, 2024), meaning that consumer preferences change as product functionalities change (Pu et al., 2025). Consumer preference analysis is typically based on the weighting of product features for conjoint analysis and measurement. As demonstrated by conjoint analysis methods, variations in features also lead to changes in preferences (Rago et al., 2025). Moreover, a key reference datum in online product features weighting calculations is the number of reviews per feature. For example, He et al. (2024) derived overall consumer preferences by weighting the number of feature reviews based on the sentiment polarity and intensity of each review. Therefore, to further enhance the robustness and effectiveness of conjoint analysis and other preference research methods, it is particularly important to develop product feature extraction techniques that more closely reflect the consumer perspective and to integrate them with MCDM approaches for conjoint analysis of feature preferences.

Information entropy has been validated through regression significance tests as an effective measure for assessing information usefulness (Fresneda & Gefen, 2020). Furthermore, integrated learning and modeling analysis have demonstrated that information entropy serves as a robust indicator of information quality (Li et al., 2024). Its advantages include the use of unordered metrics and the capability to automatically predict information quality, thereby reducing the three biases observed in the voting method. Despite the increasing attention given to information entropy research, its application in predicting useful information when combined with machine learning binary classification remains underexplored. Therefore, this study aims to fill this gap by leveraging the predictive capabilities of information entropy within a machine learning framework to enhance the accuracy of information quality assessments.

2.3. The studies of emotion and preference extraction in online reviews

Research related to prospect theory suggests that decision-making prospects are confined to either negative or positive outcomes, rather than a mix of both (Dai et al., 2024). Moreover, many preference studies indicate that regardless of the quality of the information, neutral sentiment in reviews does not contribute to preference extraction because it remains neutral. For instance, Fischer and Mueller (2025) investigated the behavioral motivations behind information retrieval, and their findings show that in various scenarios, the preference for positive and negative information significantly influences individuals' motivations to seek information, while neutral information does not have a significant impact. Serrano-Guerrero et al. (2023) noted that the Tweets were either critical or complementary, while neutral opinions were rarely observed. This observation triggered their research where the polarities of neutral views were removed from the labels of sentiment training. The results presented that this approach greatly facilitated the development of more accurate fastText models and provided more precise results of sentiment classification. Li et al. (2023) came to a similar conclusion that the removal of neutral information was required when determining the different features of the dataset. Liu et al. (2025) proposed that the online comment count model mainly deals with positivity, negativity, and neutrality; However, these do not reflect other situations with positive and negative aspects, but the overall mood is not neutral but uncertain.

Accordingly, we further removed neutral sentiment reviews on the basis of perceived usefulness analysis and calculated weighted preferences based solely on perceived useful positive and negative reviews to add to the results accuracy for preference analysis. Table 1 highlights the differences between our research and traditional related studies.

3. Methodology

3.1. Research framework

This paper proposes an innovative hybrid method, combining machine learning with multi-criteria decision-making (MCDM) methods, to extract consumer preferences from online reviews and provide decision support. As shown in Fig. 1, the method is divided into five interconnected parts: data acquisition and preprocessing, perceived useful information prediction, product feature extraction, sentiment analysis, and preference calculation and ranking.

The process overview of Parts I to V is as follows.

Part I. Data processing

We began by using web scraping techniques to collect online reviews, loading the data with Python 3.8. The preprocessing steps included normalizing the data and removing duplicates, resulting in a refined dataset comprising 70,013 reviews.

Part II. Perceived useful information prediction

A machine learning model was constructed using information entropy to predict Consumer Perceived Useful Information (CPUI). The model's predictive ability was validated against common methods, demonstrating its superiority in identifying perceived useful information.

Part III. Product feature extraction

Feature extraction methods were applied to the dataset containing identified CPUI to extract relevant product features. These methods were compared with traditional feature extraction techniques, focusing on the accuracy and capability of CPUI-based methods in revealing hidden product attributes.

Part IV. Sentiment analysis

Utilizing the consumer-perceived useful information, Natural Language Processing (NLP) techniques (BERT-wwm-ext) are employed to analyze review content, aligning it more closely with consumer perspectives. Sentiment de-neutralization was further used to reduce errors in preference analysis. The efficiency of the NLP methods was compared with state-of-the-art (SOTA) NLP techniques.

Part V. Preference Calculation and Ranking

A self-coded software based on the Best-Worst Method (BWM) from Multi-Criteria Decision Making (MCDM) was developed. This software calculated customer preferences for each product feature using

Table 1
Differences between our research and traditional related research.

Literature	Theories and methods					
	Information entropy	Usefulness of reviews	Feature analysis	Weighted analysis	Sentiment analysis	De neutralization
Chaudhuri et al. (2024)		✓	✓		✓	
Fischer and Mueller (2025)		✓	✓		✓	
Wang et al. (2024)	✓	✓				✓
Leon et al. (2024)			✓	✓	✓	
Mladenovic et al. (2024)		✓	✓		✓	
An et al. (2025)		✓	✓	✓		
Li et al. (2024)	✓		✓		✓	
Wang et al. (2022)	✓	✓				✓
Serrano-Guerrero et al. (2023)			✓	✓		✓
Zhang et al. (2025)		✓	✓		✓	
Jiang et al. (2025)			✓		✓	✓
Dong, Yao, et al. (2025)	✓		✓	✓		
Wang et al. (2023)	✓	✓				✓
Halder (2025)			✓	✓		
Zaidan et al. (2025)			✓	✓		
Li et al. (2023)				✓	✓	✓
Pu et al. (2025)			✓	✓	✓	
Sarram and Ivey (2022)			✓		✓	✓
He et al. (2024)			✓	✓	✓	
Liu et al. (2025)	✓	✓				✓
Borghi and Ratcharak (2025)		✓	✓		✓	
Rago et al. (2025)			✓		✓	
Zhang and Niu (2024)			✓	✓	✓	
Zhao et al. (2025)		✓	✓	✓		
He et al. (2024)			✓	✓	✓	
Our method	✓	✓	✓	✓	✓	✓

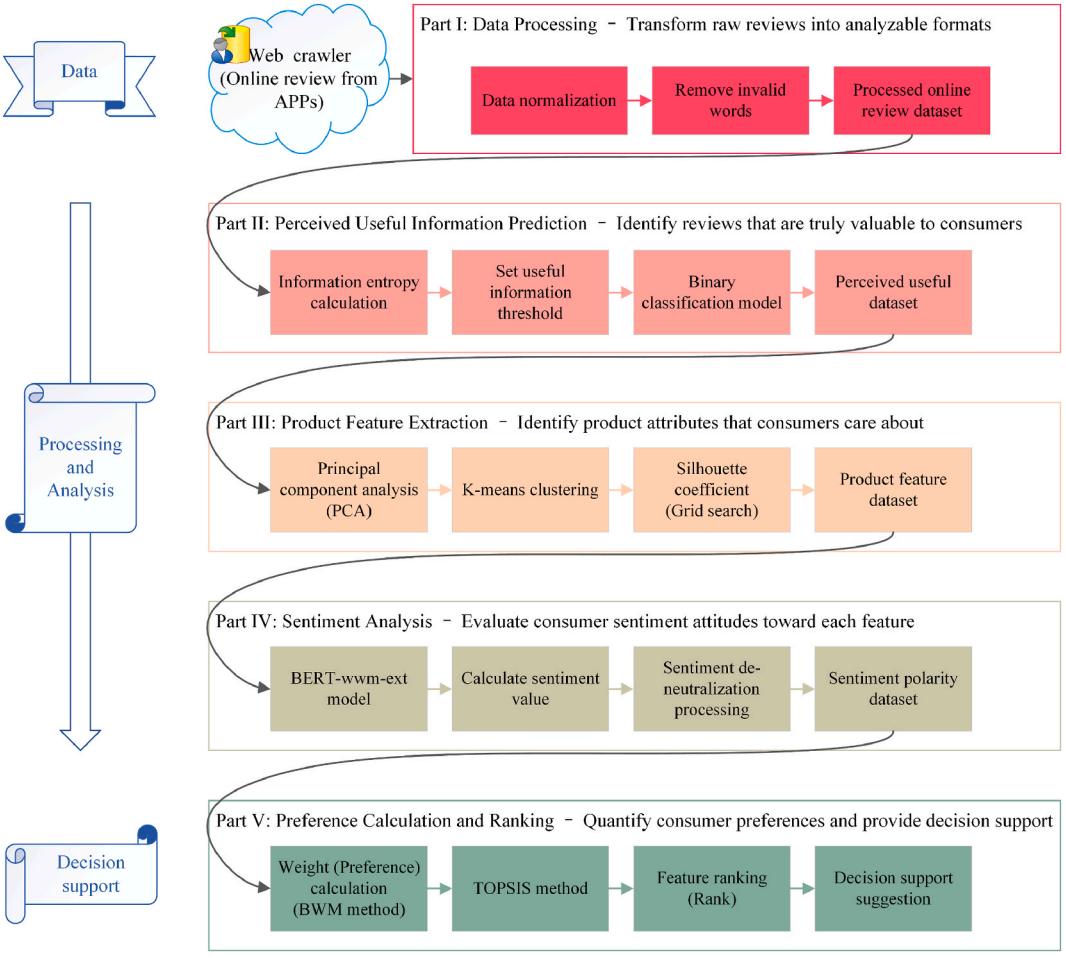


Fig. 1. Framework of hybrid machine learning and MCDM method for consumer preference extraction and decision support.

perceived usefulness information, reducing the bias of invalid information on the weighted calculation results. This new weighted calculation approach based on consumer perceived useful information provides preference extraction and analysis that is more in line with consumer perspectives, and provides more accurate decision support.

To clearly demonstrate the entire processing flow, Algorithm 1 provides the overall pseudocode for this method. The algorithm describes in detail the entire transformation process from raw online review data to the final product feature preference ranking. The following sections will elaborate on the specific implementation steps and algorithm design for each part.

Algorithm 1 Hybrid machine learning and MCDM method

Input: Online review dataset R (collected via web crawler)
Output: Feature preference ranking Rank, decision support recommendations S
//Part I: Data Preprocessing - Transform raw reviews into analyzable formats
For each review r_i in dataset R:
    Standardize r_i through deduplication, word segmentation, POS tagging, etc., to obtain r_i'
    Remove invalid words (stop words, special symbols, etc.) to get processed review r_i"
Return processed review collection R′
//Part II: Perceived Useful Information Prediction - Identify reviews that are truly
Create information entropy set E = {}
For each processed review r_i' ∈ R':
    Calculate information entropy of r_i'
    Add e_i to set E
  Calculate the average of all elements in set E:
Set useful information threshold $\tau = \mu + \alpha*\sigma$, where σ is the standard deviation of elements in E, α is an adjustment parameter
Create high information entropy review dataset R_high = {r_i' $|e_i > \tau$}
Use binary classification model to predict perceived useful reviews:
  Train binary classification model M using R_high as positive samples and the rest as negative samples
  Apply model M to predict perceived usefulness for the entire dataset, obtaining R_useful
//Part III: Product Feature Extraction - Identify product attributes that consumers care about
Use K-means and PCA combination method to cluster R_useful into K clusters
Adjust K value through grid search:
  For each candidate K value, calculate silhouette coefficient s_k
    Select optimal K value K* = argmax(s_k)
Determine product feature set F = {f_1, f_2, …, f_K*} based on the results of K* clusters
//Part IV: Sentiment Analysis - Evaluate consumer sentiment attitudes toward each feature
Create sentiment value set V = {} and sentiment polarity dataset P = {}
For reviews related to each feature f_i ∈ F:
  Calculate sentiment value v_i using BERT-wwm-ext model
  If $|v_i| > \varepsilon$ (where ε is the neutral sentiment threshold):
    Add v_i to set V
    Determine polarity p_i based on the sign of v_i, add to set P
Return V and P (elements in V and P correspond one-to-one)
//Part V: Preference Calculation and Ranking - Quantify consumer preferences and provide decision support
Create preference value set Pref = {} and weight set W = {}
Calculate the weight w_i of each feature f_i in set F using BWM method:
Use TOPSIS method to rank features, obtaining Rank
Analysis consumer preference and provide decision support recommendations S based on Rank, F, and P
Return Rank and S

In the overall framework algorithm, the data preprocessing stage first standardizes each review, including deduplication, word segmentation, and POS tagging, then removes invalid words such as stop words and special symbols, ultimately obtaining a normalized review collection. The perceived useful information prediction stage innovatively applies information entropy theory, calculates the information entropy value of each review, establishes an information entropy distribution, sets a reasonable threshold to identify high-information reviews, and trains a binary classification model based on this to predict the perceived usefulness of reviews. The product feature extraction stage combines K-means clustering with PCA dimensionality reduction, performs feature extraction on perceived useful reviews, and determines the optimal number of clusters through grid search and silhouette coefficients, thereby identifying key product features. The sentiment analysis stage uses an improved BERT-wwm-ext model to analyze the sentiment polarity of reviews, while introducing neutral sentiment threshold filtering to ensure the accuracy of subsequent preference analysis. Finally, the preference calculation and ranking stage combines BWM to determine feature weights, applies the TOPSIS method to calculate feature preference values and rankings, thereby generating the final feature preference ranking and decision support recommendations.

In the following sections, we will explain Parts I through V step by step in detail, providing clear insights into each part. Additionally, we will present pseudocode for algorithms related to specific parts, ensuring a thorough understanding of their implementation.

3.2. Data processing

Online medical products and daily necessities have always been valued and have a large customer base. As per a previous literature study on product type classification (Wang et al., 2024), we defined medical APPs as search products and toothpaste as experience products.

We collected real online review data using a web crawler program developed with Python 3.8. To ensure the efficiency and stability of data collection, we developed a distributed crawler system based on the Scrapy framework, controlling request frequency to no more than 60 times per minute, and implementing retry mechanisms and IP proxy pools to handle possible access restrictions. Additionally, for the medical and health field, we designed specialized crawling strategies to ensure the collected reviews cover diverse consumer perspectives.

The initial review data go through the following carefully designed preprocessing steps to ensure data quality.

Algorithm 2: Data Preprocessing Algorithm

Input: Raw online review dataset R_raw
Output: Processed review dataset R′
R = {}//Initialize processed review collection
For each raw review r_raw ∈ R_raw:
//Deduplication processing
    hash_value = MD5_hash(r_raw)
    If hash_value already exists in hash table:
        Continue//Skip duplicate reviews
    Else:
        Add hash_value to hash table
//Text cleaning
    r_clean = Remove HTML tags and special characters(r_raw)
//Word segmentation and POS tagging
    word_list = jieba.cut(r_clean)//Use jieba for word segmentation
    pos_list = jieba.posseg(r_clean)//POS tagging
//Filter stops words and invalid words
    filtered_words = []
    For each word and its POS tag in pos_list:
        If word not in stop word list AND pos in ['n', 'v', 'a', 'ad', 'an']://Only keep substantive words like nouns, verbs, adjectives
            filtered_words.append(word)
//Construct processed review
    If len(filtered_words) > 5://Ensure review has enough valid content
        r' = ''.join(filtered_words)
        Add r' to R
Return R

Algorithm 2 details the preprocessing workflow: first using MD5 hash algorithm for deduplication, removing 9587 duplicate reviews; then performing text cleaning, removing HTML tags and special characters; next using the "Jieba" segmenter for word segmentation and Part-of-Speech (POS) tagging, we expanded the specialized dictionary for the medical and health field, adding 2314 domain terms, significantly improving the recognition accuracy of medical terminology; subsequently filtering stop words and invalid words, referring to the Harbin Institute of Technology stop word list (Fan et al., 2025), keeping only substantive words valuable for analysis; finally, ensuring the

processed reviews contain enough valid content before adding them to the processed dataset.

3.3. Perceived useful information prediction

There is a large amount of low-value information in online reviews, directly affecting the accuracy of consumer preference extraction. To address this issue, we propose an innovative method based on information entropy and machine learning to predict perceived useful information (PUI) in reviews.

The concept of information entropy was first developed by C. E. Shannon (Shannon, 1948), who had pointed out that information entropy refers to the average amount of information after eliminating redundant information, and the greater the information entropy is, the more new information is brought in, and the higher the contribution and usefulness of information. Therefore, information entropy is proportional to the amount of information. Moreover, expressions of false reviews are more repetitive, redundant, and monotonous (Moon et al., 2019), and the advantage of information entropy enables it to solve the above problems. However, the research of information usefulness prediction based on information entropy has not been taken seriously, thus this paper proposes a novel information usefulness prediction model based on the principle of information entropy and calculates its prediction threshold, well filling the above research gap. The specific predictive modeling algorithm and logical steps are as below.

Algorithm 3: Perceived Useful Information Prediction Based on Information Entropy and Machine Learning

Input: Online review dataset, information entropy calculator, machine learning binary classifier
Output: Reviews classified as "useful" or "not useful" based on perceived usefulness
//Initialize variables
theta = Initialize model parameters()//Initialize parameters for machine learning classifier
phi = Initialize entropy parameters()//Initialize parameters for information entropy calculation
//Data preprocessing
For k in range(number of reviews):
    review = Load review(k)//Load the kth review
    review = Preprocess review(review)//Deduplicate, standardize, and tokenize review text
//Calculate information entropy
For review in reviews:
    term_probabilities = Calculate term probabilities(review)//Calculate probability of each term in the review
    information = [-log(p) for p in term_probabilities]//Calculate information for each term
    entropy = sum([p * I for p, I in zip(term_probabilities, information)])//Calculate information entropy for the entire review
//Binary classification
model = Train binary classifier(reviews, entropy, theta)//Apply binary classification model using entropy as feature
Optimize model(model, objective function, kl divergence)//Optimize theta via gradient ascent on objective function
//Model training
For epoch in range(max iterations):
    mini_batch = Sample mini batch(reviews)//Randomly sample mini-batch of training data
    theta = Update model parameters(theta, mini_batch, objective function, kl divergence)
//Update model parameters
For epoch in range(max iterations):
    beta = Adjust kl coefficient(model, validation set)//Adjust KL divergence coefficient
    If KL divergence exceeds threshold(model):
        beta = Refine sensitivity(beta)//Refine sensitivity of KL divergence coefficient
//Evaluate model performance
predictions = Classify reviews(model, reviews)//Classify reviews using trained model
TP, FP, TN, FN = Calculate confusion matrix(predictions, true_labels)
TPR, FPR, accuracy, precision, recall, f1_score, auc = Calculate performance metrics(TP, FP, TN, FN)//Calculate performance metrics

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Algorithm 3: Perceived Useful Information Prediction Based on Information Entropy and Machine Learning

//Determine optimal threshold for usefulness
optimal_threshold = Determine optimal threshold(performance_metrics)
useful_reviews = [review for review in reviews if classified as useful(review, optimal_threshold)]
Return useful_reviews, not_useful_reviews

Algorithm 3 combines information theory with machine learning through refined steps to identify useful information. The algorithm first calculates the information entropy of each review, representing the average amount of non-redundant information in the review; then uses these entropy values as features to train a binary classification model; during training, optimizes model parameters through mini-batch gradient descent and KL divergence regularization; next evaluates the model, calculating comprehensive performance metrics such as precision, recall, and F1 score; finally determines the optimal threshold based on these metrics to classify reviews as "useful" or "not useful". This approach addresses issues in traditional voting methods such as early bird bias, winner cycle bias, and imbalanced voting bias, providing a more reliable data foundation for preference analysis.

According to the steps of the algorithm, we calculated the information amount and information entropy score results of the product reviews analyzed in this paper as presented in Fig. 2, and the results of the indicators calculated by the model thresholds are shown in Table 2.

As presented in Tables 2 and it is most appropriate to set the threshold value of perceived usefulness prediction for both types of products at the mean value of information entropy. It can be interpreted that when it is below the mean, the prediction performance is poor; when it is above the mean, it is prone to misclassify useful information whose five indicators are above 0.75 as invalid, resulting in prediction errors.

We compared the performance of the useful information prediction model in this paper with the common methods in literature research that are outlined in Fig. 3.

The comparison results of our method with other perceptual useful information prediction methods for classification indicators are shown in Table 3.

We have further analyzed the indicators of the useful information prediction model for the two product types in Table 3. As shown in Fig. 4 (a) and (b) show the trend lines of indicators under different thresholds of experience type and search type respectively, (c) and (d) are the comparison plots of the indicator trends for the two product types. Based on Eqs. (1) and (2) and Fig. 4, we can see that the degree of perceived usefulness differs for different types of products with the same prediction threshold, probably because the product features of the experience type of products are more complex than the search type, thus making it necessary to consider the moderating role of product type on preferences.

3.4. Product feature extraction

We extract product features of two types of products using K-means Clustering Algorithm (K-Means) and Principal Component Analysis (PCA) in unsupervised learning. To avoid the adverse effects of noisy words, such as adverbs, we manually delete them to obtain more perfect product features. The unsupervised learning method is convenient since no training or data annotation for features or any type of product is required(Lee et al., 2020). In addition, the techniqies of unsupervised learning methods are mature and commonly used, such as dependency resolution, association rule mining, correlation, hierarchy, topic modeling, rule-based, clustering, and etc. (Kim et al., 2024). The specific

Lines	Platform	APP name	Rating	Title	Review Content	Word Segmentation	Information Entropy	Information Volume	BERT Positive	BERT Negative
0	华为应用市场	丁香医生	4		什么时候才可以看直播回放?	['才', '看', '直播', '回放']	2.0000000	8.0000000	0.7144735	0.2855265
1	华为应用市场	丁香医生	1		问病问道一半自动给我退出了，然后就登不进去了，说什么登录	['问病', '问道', '半自动', '退出', '登', '不', '进去',	3.3219281	33.2192809	0.0009504	0.9990496
2	华为应用市场	丁香医生	1		我很烦默认为公开问题。公开才应该提示！！！而不是不公开怎	['很烦', '默认', '公开', '问题', '公开', '才', '应该',	5.4039981	258.3575843	0.0010077	0.9989923
3	华为应用市场	丁香医生	1		1：个人认为医生诊断有模板2：有时限的三个问题3：发货慢4：	['1', '个人', '认为', '医生', '诊断', '模板', '2', '时	4.2479275	80.7106228	0.0018671	0.9981329
4	华为应用市场	丁香医生	1		药品明显比市价贵，发货速度慢，出现问题不处理，没有投诉渠	['药品', '明显', '市价', '贵', '发货', '速度慢', '出现	3.7004397	48.1057163	0.0009546	0.9990454
5	华为应用市场	丁香医生	5		千万别下载 里面问问题要钱 但是没关系 我可以交钱 但是回	['千万别', '下载', '里面', '问', '问题', '要钱', '没关	4.4441879	105.7335629	0.0008628	0.9991372
6	华为应用市场	丁香医生	1		只能说太坑，有那功夫还真不如去医院挂个号，好好问问医生，	['只能', '说太坑', '功夫', '还', '真', '去', '医院', '	4.0000000	64.0000000	0.0011093	0.9988907
7	华为应用市场	丁香医生	1		先说说医生，收了钱回复一个我不在，你等候着，再说下平台，	['先说', '说', '医生', '收', '钱', '回复', '不', '等候	3.7735573	53.6964683	0.0011671	0.9988329
8	华为应用市场	丁香医生	5		最真实的一面，一般人的福音醫生	['最真实', '一面', '人', '福音', '醫生']	2.3219281	11.6096405	0.9999441	0.0000559
9	华为应用市场	丁香医生	3		怎么现在打开不了 显示危险访问 联系客服	['现在', '打开', '不了', '显示', '危险', '访问', '联系	3.0000000	24.0000000	0.0009345	0.9990655
10	华为应用市场	丁香医生	5		非常涨知识，使用性很强，有个建议就是希望能有收藏的功能，	['非常', '涨', '知识', '使用', '性', '很', '强', '有个	4.7890155	135.0234679	0.9999681	0.0000319
11	华为应用市场	丁香医生	1		这款APP系统差，赠送的优惠券根本用不了！！！	['这款', 'APP', '系统', '差', '赠送', '优惠券', '根本'	3.0000000	24.0000000	0.0007960	0.9992040
12	华为应用市场	丁香医生	1		不是，我一个广州的，你快递药物从武汉、石家庄寄过来？过年	['不是', '广州', '快递', '药物', '武汉', '石家庄', '寄	4.6808134	122.6270751	0.0011431	0.9988569
13	华为应用市场	丁香医生	5		很棒。个人认为应该把它作为科普类软件，有病的人还是应该去	['很棒', '个人', '认为', '应该', '科普', '类', '软件',	3.5465936	43.4052766	0.9999659	0.0000341
14	华为应用市场	丁香医生	5		很棒	['很棒']	0.0000000	0.0000000	0.9999676	0.0000324
15	华为应用市场	丁香医生	2		挂号费骗贵，医生也很多是小地方的...可信度不确定。	['挂号费', '骗贵', '医生', '很多', '小', '地方', '可信	3.1699250	28.5293250	0.0041076	0.9958924
16	华为应用市场	丁香医生	1		药品仓库分拣太不合理了，我在南方沿海，说好了就近发货，在	['药品', '仓库', '分拣', '太', '不合理', '南方', '沿海	6.2600681	536.4359546	0.0010830	0.9989170
17	华为应用市场	丁香医生	1		我使用一次之后点进去就闪退，卸载重装之后还是这样	['使用', '一次', '之后', '点', '进去', '闪退', '卸载',	2.9477028	24.3594000	0.0007937	0.9992062
18	华为应用市场	丁香医生	5		太棒了，我们需要这样一款权威性的科普app帮助辟谣辨伪，更	['太棒了', '需要', '一款', '权威性', '科普', 'app', '有	3.9068906	58.6033589	0.9999639	0.0000361
19	华为应用市场	丁香医生	1		为什么APP里面一改昵称就闪退，很影响我的体验	['APP', '里面', '一改', '昵称', '闪退', '很', '影响',	3.0000000	24.0000000	0.0010377	0.9989623
20	华为应用市场	丁香医生	5		非常值得推荐的软件，比如其中的问诊功能，能够到这里来的医	['非常', '值得', '推荐', '软件', '问诊', '功能', '能够	3.9976703	70.6298052	0.9999667	0.0000333
21	华为应用市场	丁香医生	1		真的很坑爹啊，10块钱提问的全部是模板回复，医生都不过脑子	['真的', '很', '坑', '爹', '10', '块钱', '提问', '全部	4.0588139	69.8887250	0.0008662	0.9991339
22	华为应用市场	丁香医生	3		希望能有私人医生服务	['希望', '私人', '医生', '服务']	2.0000000	8.0000000	0.9999645	0.0000355
23	华为应用市场	丁香医生	5		很喜欢丁香医生。有问题，解答方便快捷，详细，可以解除很多	['很', '喜欢', '丁香', '医生', '问题', '解答', '方便快	3.8073549	53.3029689	0.9999660	0.0000340
24	华为应用市场	丁香医生	5		小毛病不用去医院，轻松问诊，感谢丁香医生	['小毛病', '不用', '去', '医院', '轻松', '问诊', '感谢	3.1699250	28.5293250	0.9999652	0.0000348
25	华为应用市场	丁香医生	1		暗搓搓涨价，特惠义诊比去医院挂号还贵，呵呵哒！	['暗搓', '搓', '涨价', '特惠', '义诊', '比去', '医院',	3.4594316	38.0537478	0.0009180	0.9990821
26	苹果	丁香医生	5	生活好助	生活好助手有什么事就查 放心 方便	['生活', '好助手', '事就查', '放心', '方便']	2.3219281	11.6096405	0.9999620	0.0000380
27	苹果	丁香医生	1	运营方式	运营方式丁香医生这个软件不敢否认，可以帮助到很多不懂的人	['运营', '方式', '丁香', '医生', '软件', '不敢', '否认	5.5081605	271.1680738	0.0490132	0.9509868
28	苹果	丁香医生	5	医疗知识	医疗知识普及的好APP大部分人都是医疗小白，基本上每个人对	['医疗', '知识', '普及', '好', 'APP', '大部分', '人',	4.8719281	155.9797709	0.9999601	0.0000399
29	苹果	丁香医生	5	咋不能横	咋不能横屏呢要横屏呀	['不能', '横屏', '横屏']	0.9182958	2.1699250	0.0019890	0.9980111
30	苹果	丁香医生	5	方便好用	方便好用赞👍！你说你的话	['方便', '好用', '赞', '👍', '说']	2.3219281	11.6096405	0.9999697	0.0000302
31	苹果	丁香医生	5	解决了燃	解决了燃眉之急之前就有听说过丁香医生，这次使用体验不错，	['解决', '燃眉之急', '之前', '听说', '丁香', '医生', '	3.6250000	49.0000000	0.9999660	0.0000339
32	苹果	丁香医生	5	回答专业	回答专业，没有广告偶然通过朋友推荐下载了丁香医生App。在	['回答', '专业', '没有', '广告', '偶然', '朋友', '推荐	5.1682958	202.2285750	0.9999640	0.0000360

Fig. 2. Display of partial data results of useful information prediction model.

Table 2
Results of five indicators under senarios with different thresholds for Search and Experience products.

Type of products	Prediction thresholds	*AUC*	*ACC*	*F1*	*Recall*	*Precision*
Search products	1	0.55911	0.53750	0.40090	0.59983	0.31707
	2	0.65739	0.62248	0.50624	0.72666	0.40000
	(threshold)	**0.80009**	**0.79650**	**0.81307**	**0.81687**	**0.81571**
	5	0.84070	0.82140	0.85005	0.77716	0.93816
	6	0.87853	0.86698	0.88873	0.83643	0.94814
Experience products	1	0.61181	0.59167	0.49319	0.66370	0.39837
	2	0.67716	0.67500	0.67444	0.71053	0.64286
	(threshold)	**0.79326**	**0.79167**	**0.82600**	**0.79395**	**0.86232**
	5	0.82578	0.80900	0.84480	0.77709	0.92593
	6	0.86342	0.84645	0.87628	0.81391	0.94928

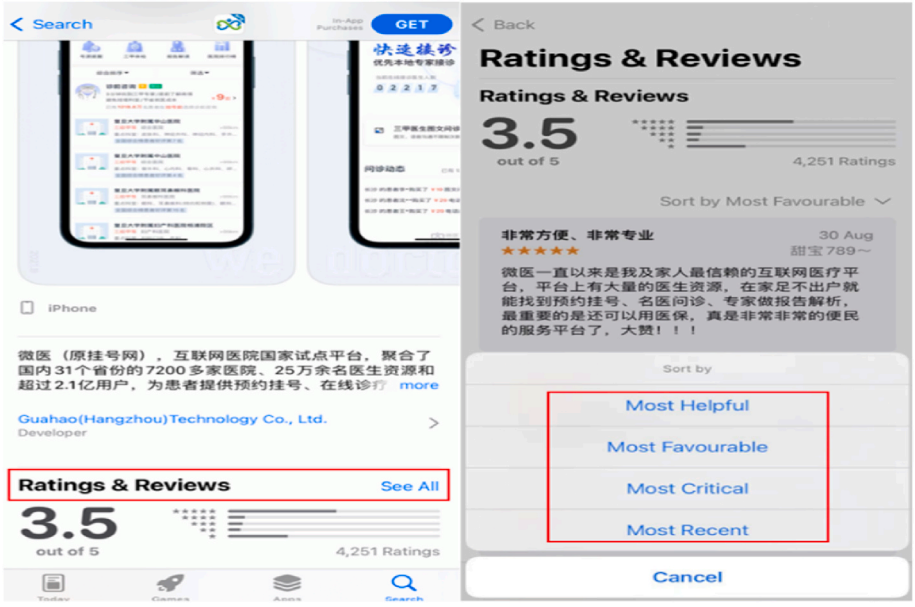


Fig. 3. Common methods of perceived usefulness prediction.

Table 3
Results of five indicators under different methods for search and experience type of products.

Type of products	Indicators	Methods				
		The most helpful	The most favorable	The most critical	The most recent	Our method
Search products	*AUC*	0.689317	0.584487	0.614167	0.560912	**0.794279**
	ACC	0.766667	0.633333	0.661111	0.611111	**0.844444**
	Precision	0.844444	0.659259	0.681481	0.644444	**0.896296**
	Recall	0.844899	0.816316	0.836407	0.798929	**0.896099**
	F1	0.844393	0.729399	0.748705	0.713023	**0.896123**
Experience Products	*AUC*	0.673693	0.591945	0.611378	0.529670	**0.806299**
	ACC	0.691667	0.591667	0.637500	0.541667	**0.787500**
	Precision	0.840000	0.573333	0.720000	0.564103	**0.940000**
	Recall	0.716189	0.717111	0.706626	0.676445	**0.770492**
	F1	0.772511	0.636742	0.712937	0.614298	**0.846847**

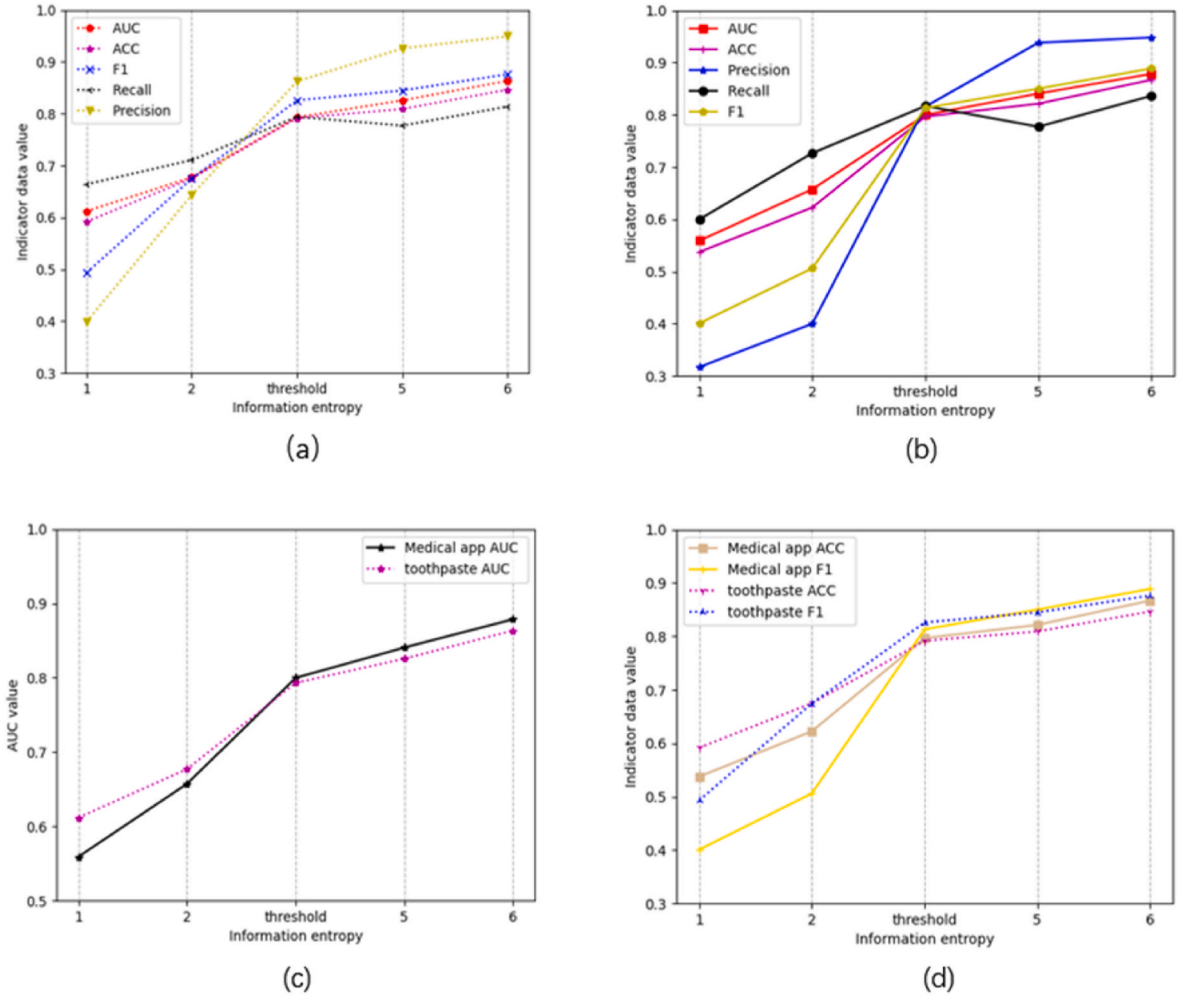


Fig. 4. Comparison of indicators for two product types.

steps of feature extractions are elaborated as below.

Algorithm 4: K-means-PCA Feature Extraction

Input: Raw review data
Output: Principal component matrix with extracted features
//Preprocess data
M = Convert to word vectors(review_data)
M = Remove noise words(M)
For row in M:
    row = row - mean(row)
M_standardized = M//Get standardized matrix M', each row has zero mean
//Calculate covariance matrix
C = (1/(n - 1)) * np.dot(M_standardized, M_standardized.T)

(*continued on next column*)

(*continued*)

Algorithm 4: K-means-PCA Feature Extraction

//Eigen decomposition
eigenvalues, eigenvectors = np.linalg.eig(C)//Perform eigen decomposition on C
D = np.diag(eigenvalues)//Convert C to a diagonal matrix D with eigenvalues on the diagonal
//Construct transformation matrix
contribution_rate = eigenvalues/np.sum(eigenvalues)//Calculate contribution rate
top_k_indices = np.argsort(contribution_rate)[-k:]//Select top k eigenvalues
T = eigenvectors[:, top_k_indices]//Construct new transformation matrix T
P = np.dot(T, M_standardized)//Get new principal component matrix P

(*continued on next page*)

(*continued*)

Algorithm 4: K-means-PCA Feature Extraction

```
//Apply K-means clustering
kmeans = KMeans(n_clusters = k)//Initialize cluster count k
kmeans.fit(P)//Cluster the principal component matrix P
centroids = kmeans.cluster_centers_//Identify center point of each cluster
Return centroids//Return K-means cluster centers, representing extracted product features
```

Algorithm 4 combines PCA dimensionality reduction and K-means clustering. The algorithm first converts reviews to word vectors and removes noise words; then standardizes the data, giving each row a zero mean; next calculates the covariance matrix and performs eigen decomposition to obtain eigenvalues and eigenvectors; based on eigenvalue contribution rates, selects the k most important eigenvalues to construct a transformation matrix; multiplies the transformation matrix with the standardized data to obtain a dimensionally reduced principal component matrix; finally applies the K-means algorithm to cluster the dimensionally reduced data, and identifies key product features through cluster centers. Compared to traditional methods, this algorithm applies PCA dimensionality reduction before clustering, not only reducing computational complexity but also improving clustering quality and feature extraction accuracy through noise reduction.

In addition, to determine the optimal number of clusters K, we conducted a grid search and used the Silhouette Coefficient to evaluate clustering quality. Table 4 shows the grid search results for hyperparameter K. Based on the maximum Silhouette Coefficient, we determined 6 main feature categories for experience products (Silhouette Coefficient 0.30297744) and 5 main feature categories for search products (Silhouette Coefficient 0.33869758). This data-driven K value selection method avoids biases that might be introduced by subjective decisions, ensuring the objectivity and scientific nature of clustering results.

Through the above method steps, we obtain the feature clustering diagram as shown in Fig. 5: (a) is the feature clustering diagram of the original reviews of search products, (b) is the feature clustering diagram of high information entropy reviews of search products, (c) is the feature clustering diagram of original reviews of experience products, and (d) is the feature clustering diagram of high information entropy reviews of experience products.

3.5. Sentiment analysis

Given that this study is about preference analysis, it is imperative to accurately classify positive and negative reviews and remove neutral sentiment reviews that cannot be analyzed for preference. The method in this paper for the sentiment analysis model outperforms Bidirectional Encoder Representation from Transformers (BERT), the SOTA baseline model in NLP sentiment analysis.

3.5.1. BERT-wwm-ext model

BERT sentiment analysis model has achieved SOTA results on 11 NLP tasks(Zhao et al., 2022). The BERT-wwm-ext method in this paper is an improvement on the BERT model. The model is constructed from the encoder of the transformer model as shown in Fig. 6. We can see from the figure that the internal structure of the transformer is a stack of Self-Attention Layer and Layer Normalization.

Table 4
Grid search results of hyperparameter *k*.

Grid search range of *k* for *Experience product*	Silhouette Coefficient of *Experience product*	Grid search range of *k* for *Search product*	Silhouette Coefficient of *Search product*
3	0.23137015	3	0.32180953
4	0.25271740	4	0.30728534
5	0.29026008	**5**	**0.33869758**
6	**0.30297744**	6	0.32710320
7	0.27424960	7	0.29810378
8	0.26726997	8	0.28823993
9	0.26743470	9	0.27153626

3.5.2. Data training and parameter set of the model

In order to avoid overfitting and improve the generalization ability of the model, we used the common 5-fold cross-validation for cross-validation. The parameter setting process and training process of the sentiment model in this paper are as follows.

Algorithm 5 BERT-wwm-ext

```
Input: Corpus of text data
Output: Trained BERT-wwm-ext model
//Data preprocessing
tokens = Tokenize corpus(corpus)//Tokenize input corpus using BERT's tokenizer
For sequence in tokens:
    sequence = Apply whole word masking(sequence, mask_prob = 0.15)//Apply whole word masking to tokens in each sequence
embeddings = Convert to embeddings(tokens)//Convert tokens to token embeddings, supplemented with segment and position embeddings
embeddings = Pad sequences(embeddings, maxlen = 512, pad_token = '[PAD]')//Use sequence length of 512, padding as necessary
embeddings = Add special tokens(embeddings)//Use [CLS] to start sequences, [SEP] to separate sentences within sequences
//Initialize model parameters
model = load_bert_base_model();
model.num_hidden_layers = 3//Set number of hidden layers
model.hidden_size = 1024//Set hidden layer size
model.pooler_num_fc_layers = 3//Set number of pooling layers
model.pooler_size_per_head = 128//Set pooling size per head
model.vocab_size = 21128//Set vocabulary size
//Define loss functions
mlm_loss = Set MLM loss()//Set masked language model loss for predicting masked tokens
nsp_loss = Set NSP loss(optional = True)//Optionally set next sentence prediction loss
//Set training parameters
optimizer = Select optimizer(Adam, lr = 5e-5)//Select Adam optimizer with learning rate 5e-5
batch_size = 16//Set batch size to 16 to optimize GPU usage
num_epochs = Specify training epochs()//Set number of training epochs
eval_intervals = Specify evaluation intervals()//Set evaluation intervals
//5-fold cross-validation
dataset = Prepare data(corpus)//Prepare data by tokenizing and splitting corpus
folds = Stratified sampling(dataset, num_folds = 5)//Stratify and randomly sample dataset into 5 folds
metrics = Initialize metrics(['Precision', 'Recall', 'F1', 'AUC', 'Accuracy'])//Initialize evaluation metrics
//Model evaluation loop
For fold in range(5):
    train_set, val_set = Get training validation sets(folds, fold)//Use one fold as validation set, rest as training set
    model = Train model(model, train_set, mlm_loss, nsp_loss, optimizer, num_epochs)//Train model
    results = Evaluate model(model, val_set)//Evaluate model on validation set
    Record metrics(metrics, results, model)//Calculate and record evaluation metrics
//Calculate statistical measures
For metric in metrics:
    mean, std_dev = Calculate mean std dev(metrics[metric])//Calculate mean and standard deviation of metric
    se = std_dev/sqrt(5)//Calculate standard error
    ci_lower = mean - 1.96 * se//Calculate confidence interval lower bound
    ci_upper = mean + 1.96 * se//Calculate confidence interval upper bound
    metrics[metric].update({'mean': mean, 'std_dev': std_dev, 'ci': (ci_lower, ci_upper)})//Store statistical results
//Save trained model and report results
Save model(model, 'trained_bert_wwm_ext_model')//Save trained model parameters
Report results(metrics)//Output average performance metrics and confidence intervals
Return model//Return trained model
```

Algorithm 5 describes in detail the training process of the BERT-

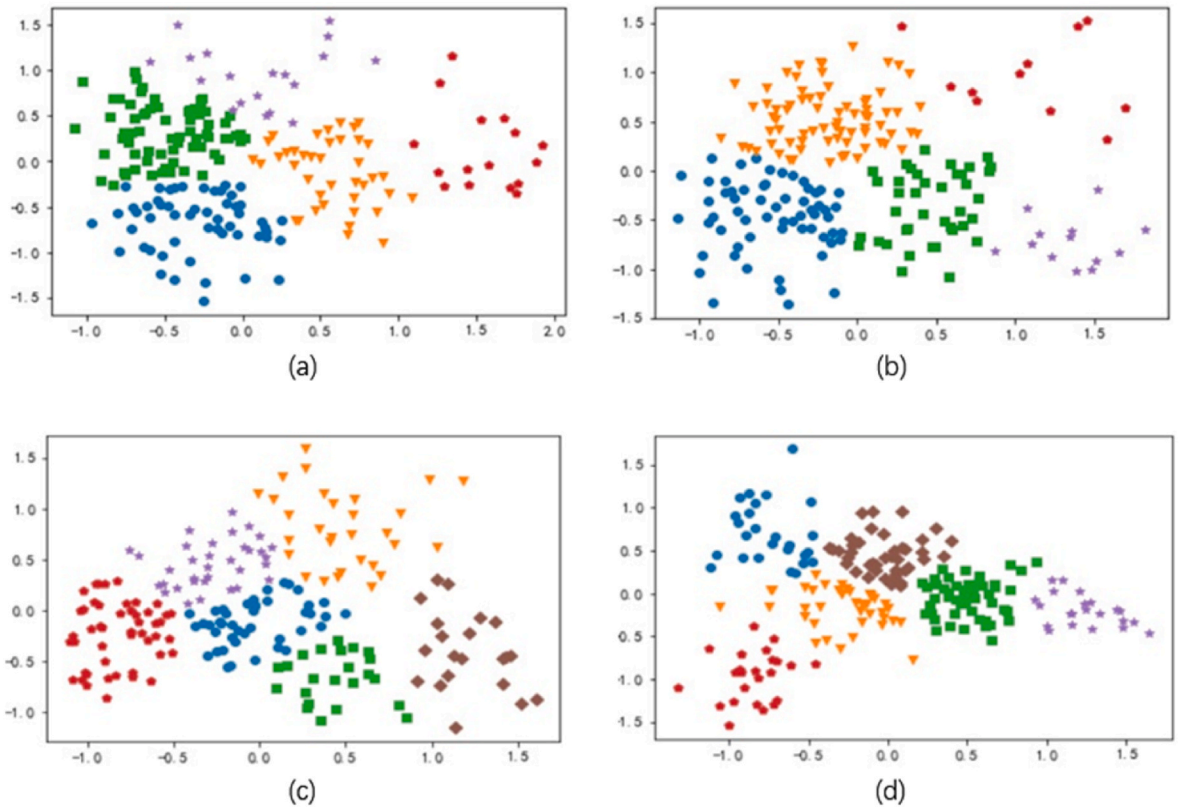


Fig. 5. The feature clustering diagrams for search product and experience product.

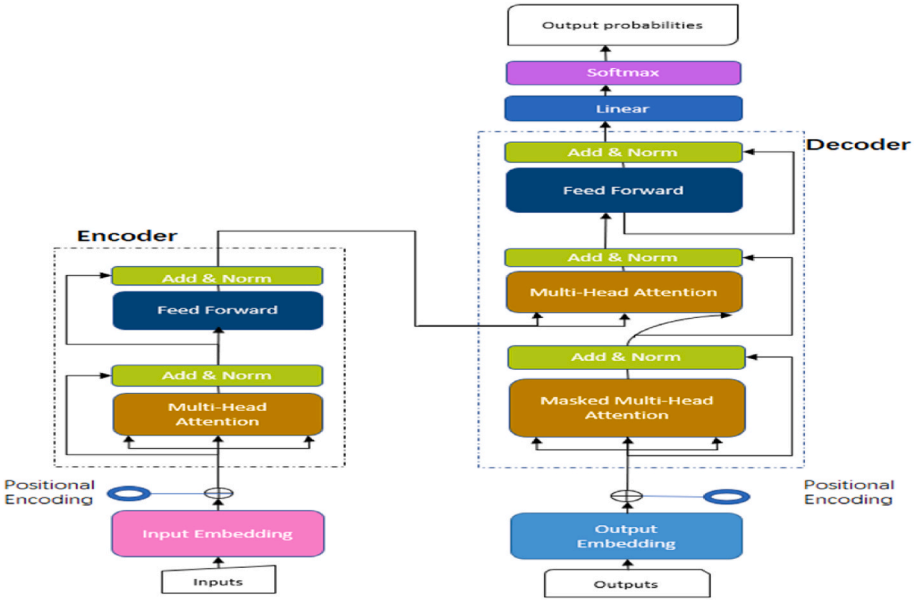


Fig. 6. Structure schematic.

wwm-ext model. We carefully tuned the model structure and parameters to adapt to the characteristics of consumer reviews. The data preprocessing stage specifically uses whole word masking technology, which is crucial for word meaning understanding in Chinese contexts; model parameter settings optimize the number, size, and structure of hidden layers to balance performance and computational complexity; the training process adopts 5-fold cross-validation to ensure the reliability and stability of results; finally, detailed statistical metrics are calculated, including mean, standard deviation, and confidence intervals, providing strict statistical guarantees for model performance. Additionally, we also introduced sentiment de-neutralization processing, removing neutral sentiment reviews, further improving the accuracy of preference analysis.

The results of cross validation are shown in Table 5, It can be seen that the selected model in this article has more advantages.

In addition, to further test the accuracy of the sentiment classification model, we compared the BERT wwm ext sentiment classification method with other commonly used methods (including SOTA methods),

Table 5
Results of Cross validation.

Type of products	model	mean	std	upper	lower
Experience Products	LogisticRegression	0.8385	0.0088	0.8386	0.8384
	SVM	0.8381	0.0180	0.8382	0.8380
	Decision tree	0.8525	0.0118	0.8526	0.8524
	KNN	0.8532	0.0120	0.8533	0.8531
	XgBoost	0.8490	0.0170	0.8491	0.8489
	BERT	0.8517	0.0063	0.8517	0.8517
	BERT-wmm	0.8647	0.0102	0.8648	0.8646
	BERT-wwm-ext	**0.8701**	**0.0130**	**0.8702**	**0.8700**
	GaussianNB	0.7811	0.0131	0.7812	0.7810
Search products	LogisticRegression	0.765	0.0154	0.7652	0.7648
	SVM	0.6481	0.0164	0.6483	0.6480
	Decision tree	0.7439	0.0133	0.7440	0.7438
	KNN	0.5552	0.0171	0.5554	0.5550
	XgBoost	0.8737	0.0112	0.8738	0.8736
	BERT	0.5798	0.0101	0.5799	0.5797
	BERT-wmm	0.9096	0.0069	0.9097	0.9095
	BERT-wwm-ext	**0.9226**	**0.0072**	**0.9227**	**0.9225**
	GaussianNB	0.5621	0.0116	0.5622	0.5620

using five common machine learning classification metrics based on the sample data in this paper. The comparison results are shown in Table 6.

3.6. TOPSIS evaluation strategy based on BWM

After completing sentiment analysis, consumer preferences for different product features need to be quantified and ranked. This section introduces our preference calculation and ranking method that combines BWM and TOPSIS.

The BWM method from MCDM is used in this paper for the calculation of feature's weight, and the weights can be determined by solving the following optimization model:

$$\min\max_j \left\{ \left| \frac{w_B}{w_j} - a_{Bj} \right|, \left| \frac{w_j}{w_W} - a_{jW} \right| \right\}$$

$$s.t. \quad \sum_j w_j = 1 \,; w_j \geq 0, j = 1, \ldots, n$$

Here, a_{Bj} denotes the degree to which a consumer prefers optimal feature B relative to the feature j, a_{jW} represents the degree to which a consumer prefers feature j over the worst feature W, w_B is the weight of the optimal feature B, w_W is the weight of the worst feature W, and w_j is the weight of any feature j.

Further, combining the above principles with Python 3.8, we have self-programed the calculation software of BWM. To verify the accuracy of our software, we run it to simulate the results based on the contents of the paper that first proposed BWM (Rezaei, 2015). The output results of our program are consistent with the paper results, as shown in Fig. 7.

Based on feature weights obtained from BWM, we use the TOPSIS (Technique for Order Preference by Similarity to an Ideal Solution) method to calculate preference values for each product feature and rank them. TOPSIS is a commonly used multi-criteria ranking method, with the basic idea of choosing solutions that are closest to the ideal solution and furthest from the negative ideal solution. The detailed calculation steps are as follows.

Step 1: Construct the original data matrix. Suppose that the original data matrix of m product features and n sub-product features is $F = (F_{ij})_{m\times n} = (w_{ij}{}^{*}N_{ij})_{m\times n}$, here, w_{ij} is the weight of the sub-product

Table 6
Comparison of 5 indicators between our method and other sentiment analysis methods.

Product type	Model	Indicators				
		AUC	ACC	Precision	Recall	F1
Experience Product (Original reviews)	XgBoost	0.74117	0.76994	0.90713	0.52342	0.66381
	SVM	0.76127	0.76947	0.75211	0.69919	0.72469
	Decision tree	0.75136	0.77590	0.87323	0.56567	0.68658
	Logistic Regression	0.75352	0.77987	0.90021	0.55413	0.68599
	KNN	0.77548	0.79707	0.88476	0.61207	0.72357
	GaussianNB	0.76044	0.78762	0.92618	0.55478	0.69391
	BERT	0.83781	0.85356	0.90492	0.73750	0.80993
	BERT-wmm	0.83080	0.84586	0.86244	0.78997	0.82930
	BERT-wwm-ext	**0.84300**	**0.85557**	**0.88112**	**0.79816**	**0.83450**
Experience Product (Perceived useful reviews)	XgBoost	0.55729	0.75522	0.75027	0.99397	0.85510
	SVM	0.80313	0.79382	0.92185	0.78260	0.84654
	Decision tree	0.55385	0.75230	0.74886	0.99167	0.85333
	Logistic Regression	0.53625	0.74437	0.74139	0.99540	0.84982
	KNN	0.57364	0.74750	0.75853	0.95721	0.84637
	GaussianNB	0.80603	0.73957	0.97371	0.65939	0.78630
	BERT	0.84014	0.89670	0.94805	0.76983	0.84509
	BERT-wmm	0.86313	0.88737	0.90558	0.82229	0.86446
	BERT-wwm-ext	**0.87533**	**0.89870**	**0.92426**	**0.83049**	**0.86966**
Search Product (Original reviews)	XgBoost	0.71781	0.74726	0.51011	0.65658	0.57415
	SVM	0.74329	0.86048	0.93054	0.49965	0.65018
	Decision tree	0.80221	0.89189	0.95005	0.61576	0.74722
	Logistic Regression	0.81319	0.90139	0.98460	0.62984	0.76824
	KNN	0.69134	0.83601	0.94549	0.39057	0.55279
	GaussianNB	0.72475	0.64719	0.41565	0.88600	0.56584
	BERT	0.89553	0.94412	0.98775	0.79451	0.88066
	BERT-wmm	0.91845	0.93350	0.93827	0.86580	0.90513
	BERT-wwm-ext	**0.93063**	**0.95320**	**0.95345**	**0.89398**	**0.92031**
Search Product (Perceived useful reviews)	XgBoost	0.77220	0.81116	0.47368	0.71227	0.56898
	SVM	0.69085	0.88050	0.82961	0.39910	0.53893
	Decision tree	0.77137	0.91414	0.92864	0.55172	0.69220
	Logistic Regression	0.80828	0.93095	0.97770	0.61956	0.75848
	KNN	0.73746	0.90375	0.93833	0.48163	0.63653
	GaussianNB	0.66278	0.50054	0.24800	0.91238	0.39000
	BERT	0.89067	0.96033	0.98718	0.78349	0.87362
	BERT-wmm	0.93193	0.94698	0.95176	0.87928	0.91861
	BERT-wwm-ext	**0.94413**	**0.95670**	**0.97044**	**0.88748**	**0.92381**

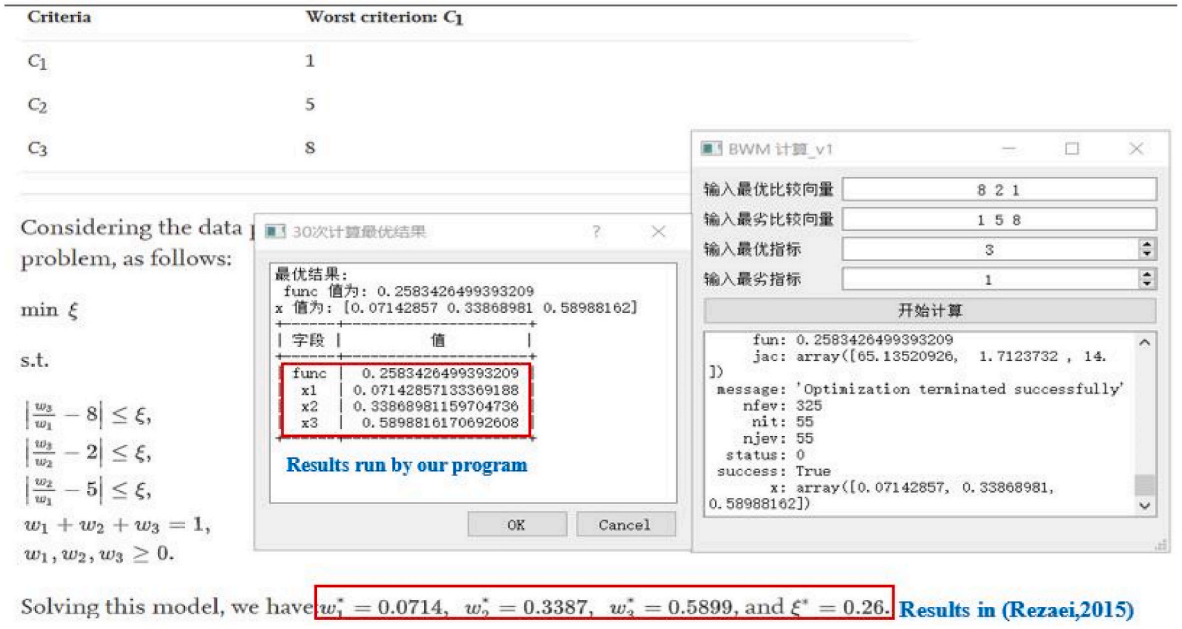


Fig. 7. Screenshot of self-programmed BWM software.

feature j for product feature i obtain by BWM, while N_{ij} is the corresponding review number;

Step 2: Matrix normalization. Let the normalized element (that is, the preference) be Z_{ij}, for any $i = 1, \ldots, m$, $j = 1, \ldots, n$ then the normalized formula is

$$Z_{ij} = \frac{F_{ij}}{\sqrt{\sum_{j=1}^{n} (F_{ij})^2}};$$

Step 3: Determine the ideal and negative-ideal solutions. The calculation formulas for the ideal solution Z_{i+} and worst solution Z_{i-} are:

$$Z_{i+} = \max_j Z_{ij}, Z_{i-} = \min_j Z_{ij};$$

Step 4: Calculate distances. Calculate distances to the ideal and negative-ideal solutions, that is the distance D_{i+} between the sub-product feature Z_{ij} and the best solution Z_{i+}, and the distance D_{i-} between the sub-product feature Z_{ij} and the worst solution Z_{i-}, as follows:

$$D_{i+} = \sqrt{\sum_{j=1}^{n} (Z_{i+} - Z_{ij})^2}, D_{i-} = \sqrt{\sum_{j=1}^{n} (Z_{i-} - Z_{ij})^2};$$

Step 5: Quantitative assessment of the degree of closeness between each feature and the optimal solution, as follows:

$$C_i = D_{i-} / (D_{i+} + D_{i-})$$

obviously, the closer the C_i is to 1, the more critical the feature is.

Step 6: Sort C_i in descending order and provide the corresponding evaluation results.

4. Case study

4.1. Product feature extraction based on CPUI

Following the approach in Sections 3.2 - 3.3, from 34,750 real online reviews of search products, 19,711 reviews were perceived to be useful, representing 56.722 % of the original reviews; from 35,263 real online reviews of experience products, 15,971 reviews were perceived to be useful, representing 45.291 % of the original reviews.

Extracting features from the predicted consumer-perceived useful reviews in previous sections can rectify the misleading of invalid information on feature mining. In addition, it is capable of mining the hidden features after the product update in real-time given the attribute of information entropy uncertainty in the prediction method of this paper. The information entropy uncertainty aforementioned means that the less inferable content, the greater the uncertainty. Applied to the domain of online review research, uncertainty refers to emerging reviews of product features. Tables 7–10 show the extraction of consumer-perceived useful features for Experience products and Search products as well as the original feature extraction.

As can be seen in Tables 7–10, our approach mines the changes in consumer attention to features of experience and search products after updates, as well as the hidden features that interest consumers. Taking the experience product as an example, the features of the packaging 'Box', the convenience to 'Uncap' toothpaste cover, and the 'Taste' of the paste in the traditional method is missing in our approach, probably because the packaging 'Box' has become more solid and it's getting easier to 'Uncap' the toothpaste with the improvement of platform regulation and product upgrading, thus these two features are not of consumer interests now. In addition, the 'Taste' is not a feature that consumers are interested in anymore. Instead, the two features of 'Repair' and the continuous experience in 'The second day' have appeared in our method, since people are more concerned about the effect features of toothpaste, such as gum repair and sterilization continuity. In addition, the new features mined according to the uncertainty of information entropy show that the recent 'Set' and 'Freebie' activities have attracted consumers to post new reviews, indicating that they have attracted new attention. Likewise, for the search products, the software login problem has been improved after the medical app update. Hence, the perceived usefulness information analysis in this paper presents that

Table 7
Product features of Experience product (Toothpastes) of traditional method.

Price	Customer Loyalty	Effectiveness and Sustainability	Commodity	Experience	Logistics
Activity	Always	Breath	**Box**	Brushing	Delivery
Cost effectiveness	Brand	Effect	Feature	Clean	Express delivery
Discount	Customer service	For a while	Product	Comfort	Order
Supermarket	Household	Gums	Quality	Foam	Received
Worthwhile	Issue	Oral cavity	Quantity	Mouthfeel	Shipping
	Repurchase	Tooth care	Trademark	Self-supporting	Speed
	Service			Sensitive	
	The first time			Shopping	
	Trust			Stimulation	
				Taste	
				Use	
				Smell	
				Uncap	

Table 8
Product features of Experience product (Toothpastes) of our method.

Price	Customer Loyalty	Effectiveness and Sustainability	Commodity	Experience	Logistics
Activity	Always	Breath	Feature	Brushing	Delivery
Cost effectiveness	Brand	Effect	Product	Clean	Express delivery
Discount	Customer service	For a while	Quality	Comfort	Order
Freebie	Household	Gums	Quantity	Foam	Received
Set	Issue	Oral cavity	Trademark	Mouthfeel	Shipping
Supermarket	Repurchase	**Repair**		Self-supporting	Speed
Worthwhile	Service	**The second day**		Sensitive	
	The first time	Tooth care		Shopping	
	Trust			Stimulation	
				Use	
				Smell	

Table 9
Product features of Search product (Medical APPs) of traditional method.

Health Science	Medical Service	Offline activities	Online diagnosis	Software Interaction
Abundant	Body	Free	Attitude	Answer
Content	Experience	Help	Detailed	Application
Expert	Hospital	Issue	Doctor	Application Software
Friend	Line up	Redemption	Patient	Customer service
Health	Medical care	Time	Reply	Download
Knowledge	Medical examination		Timeliness	Function
Life	Service			Information
Proposal	Settle			Interface
	Solve			**Login**
	Understand			Mobile Phone
				Online
				Platform
				Running
				Speed
				Update
				User
				Version

Table 10
Product features of Search product (Medical APPs) of our method.

Health Science	Medical Service	Offline activities	Online diagnosis	Software Interaction
Abundant	Body	Free	Attitude	Answer
Content	Experience	Help	Detailed	Application
Expert	Hospital	Issue	Doctor	Application Software
Friend	Line up	Redemption	Patient	Customer service
Health	Medical care	Time	Reply	Download
illness	Medical examination		Timeliness	Function
Knowledge	Service			Information
Life	Settle			Interface
Proposal	Solve			Mobile Phone
	Understand			Online
				Platform
				Running
				Speed
				Update
				User
				Version

'Login' is no longer a feature that consumers are interested in currently. While the mining of our method shows that users care more about the descriptions of 'illness', because the more detailed the descriptions of 'illness', the more secure patients may feel.

4.2. Rectification on base data for product weighting calculation by our method

Useful information can also be called CPUI, and the number of product feature reviews changes with invalid information removed. In online customer reviews, the number of feature reviews is the most important reference data for calculating weights, also known as preferences. Tables 11 and 12 overall present the product features extracted under the traditional method and our method, illustrating the rectification of our method on the number of extracted product features. All reviews with useless neutral sentiment for preference analysis are removed from the analysis by the method in Section 3.6. It provides a solid data basis for the weighted calculation of perceived usefulness.

Table 11
The Quantity of positive and negative reviews comparison for Experience product (Toothpastes).

Product features	Traditional Method						Our Method					
	Sub product features	Quantity of Positive reviews	Quantity of Negative reviews	Sub product features	Quantity of Positive reviews	Quantity of Negative reviews	Sub product features	Quantity of Positive reviews	Quantity of Negative reviews	Sub product features	Quantity of Positive reviews	Quantity of Negative reviews
Price	Activity	2689	431	Supermarket	2054	656	Activity	2366	315	**Freebie**	256	229
	Cost effectiveness	894	25	Worthwhile	1675	59	Cost effectiveness	689	17	**Set**	534	35
	Discount	1031	233				Discount	876	160	Supermarket	1227	330
										Worthwhile	1391	45
Customer Loyalty	Always	3623	562	Repurchase	802	109	Always	2858	404	Repurchase	729	82
	Brand	1324	289	Service	888	301	Brand	1111	199	Service	742	183
	Customer service	129	491	The first time	658	384	Customer service	115	353	The first time	559	238
	Household	865	89	Trust	1042	19	Household	826	76	Trust	895	19
	Issue	767	339				Issue	727	240			
Effectiveness and Sustainability	Breath	1643	133	Oral cavity	1455	173	Breath	1542	110	Oral cavity	1394	131
	Effect	3639	834	Tooth care	653	105	Effect	2938	396	**Repair**	400	86
	For a while	729	74				For a while	645	49	**The second day**	505	124
	Gums	2005	326				Gums	1914	209	Tooth care	631	78
Commodity	**Box**	121	635	Quality	1515	443	Feature	1528	207	Quantity	484	111
	Feature	1706	324	Quantity	562	148	Product	842	203	Trademark	1615	104
	Product	1016	329	Trademark	1803	134	Quality	1119	211			
Experience	Brushing	1598	315	Shopping	818	228	Brushing	1464	211	Shopping	696	165
	Clean	531	227	Stimulation	695	77	Clean	595	61	Stimulation	662	64
	Comfort	1386	269	**Taste**	871	52	Comfort	1307	248	Use	2287	315
	Foam	1337	405	Use	2580	449	Foam	1245	213	Smell	719	46
	Mouthfeel	673	88	Smell	739	59	Mouthfeel	545	46			
	Self-supporting	596	245	**Uncap**	297	496	Self-supporting	574	173			
	Sensitive	1171	185				Sensitive	1111	143			
Logistics	Delivery	1145	303	Received	1250	1046	Delivery	883	173	Received	1031	478
	Express delivery	1552	1017	Shipping	632	211	Express delivery	1341	571	Shipping	493	134
	Order	772	303	Speed	1369	178	Order	760	246	Speed	1034	116

Table 12
The Quantity of positive and negative reviews comparison for Search product (Medical APPs).

Product features	Traditional Method						Our Method					
	Sub product features	Quantity of Positive reviews	Quantity of Negative reviews	Sub product features	Quantity of Positive reviews	Quantity of Negative reviews	Sub product features	Quantity of Positive reviews	Quantity of Negative reviews	Sub product features	Quantity of Positive reviews	Quantity of Negative reviews
Medical	Body	502	48	Medical examination	306	42	Body	440	45	Medical examination	297	37
service	Experience	285	92	Service	656	201	Experience	254	85	Service	571	182
	Hospital	2046	1044	Settle	693	382	Hospital	1848	914	Settle	591	315
	Line up	405	357	Solve	526	47	Line up	361	290	Solve	463	43
	Medical care	502	69	Understand	303	63	Medical care	454	60	Understand	267	59
Offline	Free	550	210	Redemption	162	155	Free	481	184	Redemption	113	124
activities	Help	611	104	Time	797	449	Help	488	83	Time	712	394
	Issue	1510	830				Issue	1340	701			
Health	Abundant	464	104	Health	2019	176	Abundant	403	86	Health	1704	156
Science	Content	369	83	Knowledge	636	10	Content	307	70	**illness**	264	51
	Expert	440	197	Life	544	8	Expert	381	173	Knowledge	505	9
	Friend	599	91	Proposal	493	158	Friend	530	81	Life	446	7
										Proposal	444	145
Online	Attitude	485	113	Patient	306	114	Attitude	409	92	Patient	281	94
diagnosis	Detailed	309	55	Reply	803	613	Detailed	271	45	Reply	608	474
	Doctor	1660	516	Timeliness	786	403	Doctor	1364	290	Timeliness	449	135
Experience	Answer	650	345	Mobile Phone	534	404	Answer	547	261	Online	629	354
	Application	906	172	Online	689	427	Application	691	154	Platform	707	244
	Application Software	320	65	Platform	789	272	Application Software	306	62	Running	308	49
	Customer service	61	333	Running	364	56	Customer service	59	289	Speed	508	87
	Download	829	454	Speed	683	98	Download	722	381	Update	217	340
	Function	1168	132	Update	242	516	Function	939	122	User	128	151
	Information	231	241	User	138	169	Information	186	194	Version	159	128
	Interface	383	67	Version	199	152	Interface	342	60			
	Login	279	96				Mobile Phone	509	369			

4.3. Product feature weight calculation

Consumers hold their own preferences for product features and have various concerns about different product features, whereby businesses and scholars have been paying attention to the feature weighting calculation. Tables 13 and 14 show the weights of product features of Experience products. Tables 15 and 16 show the weights of product features of Search products. As shown in Tables 13–16, in comparison with the traditional method, since the weight calculation data in this paper is what consumers perceive useful, our method reflects consumer preferences more precisely.

4.4. Results summary of the case study

The case analysis results are presented in Table 17. Compared with the traditional method, this paper uses only 56.722 % of the original data for Search products (19,711 perceived useful reviews out of the 34,750 original reviews) and 45.291 % of the original data for Experience products (15,971 perceived useful reviews out of the 35,263 original reviews) to achieve highly consistent results of feature extraction with the traditional method. The remaining small number of inconsistent features are the hidden features mined by our method, effectively solving the problem caused by information overload and improving the accuracy of results in feature-based preference analysis. In addition, the numerical results of weighted calculation for preference weights (including preference ranking) based on perceived useful data sources after removing the neutral sentiment are significantly different from the traditional method. It implies that perceived usefulness based on de-neutral sentiment has a significant rectification effect on the results since the data source is perceived useful by consumers and the resultant calculated preferences are more compatible with the consumer's point of view.

Table 14
Classification Weight under traditional and our method for Experience product (Toothpastes).

Product features	Traditional Method	Our Method
	Weight of product features	Weight of product features
Commodity	0.12642921	0.10777356
Customer Loyalty	0.18352032	0.17682499
Effectiveness and Sustainability	0.17032302	0.20626657
Experience	0.23715285	0.22820857
Logistics	0.14151104	0.12603602
Price	0.14106357	0.15489027

4.5. Comparison experiment and results analysis

4.5.1. Experimental design and comparison methods

To systematically evaluate the effectiveness of the method proposed in this paper, we developed a comparative experimental framework. Nine algorithms from the literatures in the field of sentiment analysis were selected as benchmarks for comparison. The experiments are conducted also on the two distinct datasets: an experience product dataset and a search product dataset. This design ensures a rigorous evaluation of the model's cross-domain adaptability.

The evaluation framework incorporates five complementary metrics: AUC, ACC, F1, Precision, and Recall, providing a well-rounded standard for performance assessment. The comparative methods used include.

(1) RoBERTa (Deng & Liu, 2025) [1]: An optimized pre-trained language model based on BERT, designed with dynamic masking and larger-scale training, which significantly enhances text representation capabilities.

Table 13
The Weight of Product features under traditional and our method for Experience product (Toothpastes).

Product features	Traditional Method				Our Method			
	Sub product features	Weight of Product feature	Sub product features	Weight of Product feature	Sub product features	Weight of Product feature	Sub product features	Weight of Product feature
Price	Activity	0.04515604	Supermarket	0.03922053	Activity	0.05029076	**Freebie**	0.00849077
	Cost effectiveness	0.01330029	Worthwhile	0.02509458	Cost effectiveness	0.01198942	**Set**	0.00944736
	Discount	0.01829213			Discount	0.01633748	Supermarket	0.03240045
							Worthwhile	0.02593403
Customer Loyalty	Always	0.06056671	Repurchase	0.01318441	Always	0.06077332	Repurchase	0.01311579
	Brand	0.02334320	Service	0.01720687	Brand	0.02118405	Service	0.01479017
	Customer service	0.00897294	The first time	0.01507993	Customer service	0.00874213	The first time	0.01290629
	Household	0.01380626	Trust	0.01535410	Household	0.01445456	Trust	0.01497813
	Issue	0.01600590			Issue	0.01588055		
Effectiveness and Sustainability	Breath	0.02570245	Oral cavity	0.02356013	Breath	0.03031803	Oral cavity	0.03180584
	Effect	0.06473394	Tooth care	0.01096989	Effect	0.05889937	**Repair**	0.00839348
	For a while	0.01162128			For a while	0.01136477	**The second day**	0.01046189
	Gums	0.03373533			Gums	0.04297022	Tooth care	0.01205297
Commodity	**Box**	0.01094051	Quality	0.02833573	Feature	0.02918349	Quantity	0.00983897
	Feature	0.02938123	Quantity	0.01027501	Product	0.01764826	Trademark	0.02959234
	Product	0.01946409	Trademark	0.02803264	Quality	0.02151050		
Experience	Brushing	0.02768566	Shopping	0.01513708	Brushing	0.03121753	Shopping	0.01405273
	Clean	0.01096990	Stimulation	0.01117240	Clean	0.01078275	Stimulation	0.01187582
	Comfort	0.02395185	**Taste**	0.01335764	Comfort	0.02681398	Use	0.05013603
	Foam	0.02520976	Use	0.04383554	Foam	0.02876584	Smell	0.01248129
	Mouthfeel	0.01101321	Smell	0.01154867	Mouthfeel	0.00978492		
	Self-supporting	0.01217062	**Uncap**	0.01147670	Self-supporting	0.01224882		
	Sensitive	0.01962382			Sensitive	0.02004886		
Logistics	Delivery	0.02095466	Received	0.03323052	Delivery	0.01750530	Received	0.03150926
	Express delivery	0.03718098	Shipping	0.01219971	Express delivery	0.03255694	Shipping	0.01033973
	Order	0.01555693	Speed	0.02238824	Order	0.01594498	Speed	0.01817981

Table 15
The Weight of Product features under traditional and our method for Search product (Medical APPs).

Product features	Traditional Method				Our Method			
	Sub product features	Weight of Product feature	Sub product features	Weight of Product feature	Sub product features	Weight of Product feature	Sub product features	Weight of Product feature
Medical	Body	0.01411524	Medical examination	0.00893109	Body	0.01484451	Medical examination	0.01022282
service	Experience	0.00967535	Service	0.02199405	Experience	0.01037586	Service	0.02304726
	Hospital	0.07930191	Settle	0.02758892	Hospital	0.08453721	Settle	0.02773018
	Line up	0.01955603	Solve	0.01470549	Line up	0.01992532	Solve	0.01548726
	Medical care	0.01465419	Understand	0.00939303	Medical care	0.01573213	Understand	0.00997796
Offline activities	Free	0.01950469	Redemption	0.00813551	Free	0.02035382	Redemption	0.00725392
	Help	0.01834980	Time	0.03197742	Help	0.01747674	Time	0.03385162
	Issue	0.06005389			Issue	0.06246939		
Health Science	Abundant	0.01457718	Health	0.05633261	Abundant	0.01496694	Health	0.05692948
	Content	0.01160015	Knowledge	0.01657898	Content	0.01153893	**illness**	0.00964128
	Expert	0.01634800	Life	0.01416656	Expert	0.01695642	Knowledge	0.01573213
	Friend	0.01770820	Proposal	0.01670730	Friend	0.01870103	Life	0.01386508
							Proposal	0.01802767
Online diagnosis	Attitude	0.01534711	Patient	0.01077890	Attitude	0.01533423	Patient	0.01147772
	Detailed	0.00934172	Reply	0.03634030	Detailed	0.00967189	Reply	0.03311704
	Doctor	0.05584499	Timeliness	0.03051456	Doctor	0.05062439	Timeliness	0.01787463
Software interaction	Answer	0.02553523	Mobile Phone	0.02407226	Answer	0.02473066	Mobile Phone	0.02687316
	Application	0.02766702	Online	0.02864123	Application	0.02586313	Online	0.03008692
	Application Software	0.00988068	Platform	0.02722932	Application Software	0.01126347	Platform	0.02910749
	Customer service	0.01011161	Running	0.01077838	Customer service	0.01065132	Running	0.01092679
	Download	0.03292793	Speed	0.02004432	Download	0.03375979	Speed	0.01821132
	Function	0.03336258	Update	0.01945457	Function	0.03247429	Update	0.01704824
	Information	0.01211255	User	0.00787896	Information	0.01163075	User	0.00853942
	Interface	0.01154887	Version	0.00900786	Interface	0.01230411	Version	0.00878428
	Login	0.00962344						

Table 16
Classification Weight under traditional and our method for Search product (Medical APPs).

Product features	Traditional Method	Our Method
	Weight of product features	Weight of product features
Health Science	0.16401899	0.17635896
Medical Service	0.21991531	0.23188051
Offline activities	0.13802130	0.14140548
Online diagnosis	0.15816759	0.13809990
Software Interaction	0.31987680	0.31225514

(2) ALiteBERT (Jin et al., 2025) [2]: A lightweight BERT variant developed via knowledge distillation and parameter pruning, enabling a substantial reduction in computational overhead while maintaining high performance.
(3) XLNet (Hashmi & Yayilgan, 2024) [3]: Introduces permutation language modeling and Transformer-XL techniques, effectively addressing the limitation of independently masked tokens in BERT. XLNet has demonstrated superior performance compared to BERT in multiple benchmark evaluations.
(4) ELECTRA + CNN (Mohawesh et al., 2024) [4]: Combines the "replaced token detection" pre-training task with the local feature extraction strengths of convolutional neural networks (CNNs), achieving improvements in computational efficiency and feature representation.
(5) SA-attention-Bi-LSTM-BERT (Li & Wang, 2025) [5]: Integrates self-attention mechanisms, bidirectional LSTM, and BERT into a multi-level architecture to enhance contextual understanding and long-range dependency modeling.
(6) XGBoost + Lexicon (Tsiourlini et al., 2024) [6]: Combines the strengths of efficient ensemble learning and sentiment lexicon knowledge to balance computational efficiency with interpretability, making it adaptable to datasets of varying sizes.
(7) PCA + K-means + Snow (Wang et al., 2023) [7]: Employs a multi-stage processing pipeline that combines dimensionality reduction, clustering, and classification, offering reliable sentiment analysis even under resource-constrained conditions.

Table 17
The summary of case study.

Product type	Product Features	Traditional method	Our method	Traditional/Our method	Traditional method	Our method
		No. of Features	No. of Features	Common features	Weight and ranking	Weight and ranking
Experience product	Commodity	6	5	5	6(0.12642921)	6(0.10777356)
	Customer Loyalty	9	9	9	2(0.18352032)	3(0.17682499)
	Effectiveness and Sustainability	6	8	6	3(0.17032302)	2(0.20626657)
	Experience	13	11	11	1(0.23715285)	1(0.22820857)
	Logistics	6	6	6	4(0.14151104)	5(0.12603602)
	Price	5	7	5	5(0.14106357)	4(0.15489027)
Search Product	Health Science	8	9	8	3(0.16401899)	3(0.17635896)
	Medical Service	10	10	10	2(0.21991531)	2(0.23188051)
	Offline activities	5	5	5	5(0.13802130)	4(0.14140548)
	Online diagnosis	6	6	6	4(0.15816759)	5(0.13809990)
	Software Interaction	17	16	16	1(0.31987680)	1(0.31225514)

(8) DC-BiLSTM-CNN (Dong, Li, et al., 2025) [8]: Features an innovative dual-channel architecture that separately processes local and global features, enabling effective multi-scale sentiment feature extraction.
(9) Ekman-BERT (Yun et al., 2024) [9]: Integrates psychological emotion theory with deep learning to establish an emotion hierarchy classification framework, significantly improving the model's precision in emotion understanding.

The experimental results for the two datasets are presented in Tables 18 and 19. In these tables, bolded values indicate the best-performing results among all comparative methods.

4.5.2. Results analysis on the experience product data

The results presented in Table 18 indicate that the proposed method significantly outperforms all comparative models on the experiential product review dataset. Specifically, our method achieves an AUC score of 0.87533, representing a 3.9 % improvement over the second-best model (Ekman-BERT). For the ACC metric, our method achieves a score of 0.89870, with an even more notable improvement. Similarly, our method demonstrates superior performance across the three key metrics: F1, Precision, and Recall. To provide an intuitive demonstration of the effectiveness of our approach, we visualized the statistical results from Table 18 in Fig. 8 through 10.

The radar chart in Fig. 8 illustrates the comprehensive performance of each model across the five evaluation metrics. As shown in the chart, our method forms the largest enclosed area compared to other models, highlighting its overall advantage. Specifically, our model improves the AUC metric by 2.1 % and the ACC metric by 2.8 % over the second-best model (Ekman-BERT) and achieves optimal performance in F1, Precision, and Recall.

Fig. 9 focuses on the F1 metric, presenting the improvement rates of our method compared to each comparative model. The improvement rate is calculated using the following formula:

$$improvement = \frac{(F1\ of\ our\ method) - (F1\ of\ any\ each\ comparison\ method)}{F1\ of\ any\ each\ comparison\ method} \times 100\%$$

The results indicate that our model achieves varying degrees of performance improvement over existing methods. The highest improvement is 7.79 % (compared to DC-BiLSTM-CNN), while the lowest improvement is 3.54 % (compared to Ekman-BERT). The average improvement is 5.5 %, calculated as the mean improvement across all methods. These data strongly demonstrate the substantial advancements achieved by our method.

Fig. 10 uses a bubble chart to visualize the trade-off relationship among Precision, Recall, and F1. In the chart, the horizontal and vertical axes represent Recall and Precision, respectively, while the bubble size is proportional to the F1. The ideal scenario is for the bubble to be the largest and positioned in the upper-right corner, where both Precision and Recall are high. Clearly, our method achieves this ideal state, exhibiting the largest bubble size and most favorable position, thereby attaining the best balance among the three metrics.

Table 18
Evaluation index values corresponding to comparison methods based on experience product online review data.

Method	AUC	ACC	F1	Precision	Recall
RoBERTa [1]	0.84321	0.83245	0.83807	0.86137	0.81564
ALiteBERT [2]	0.81234	0.80987	0.81104	0.83655	0.78932
XLNET [3]	0.83219	0.82476	0.83500	0.85794	0.81267
ELECTRA + CNN [4]	0.81655	0.80892	0.82202	0.84231	0.80246
SA-attention-Bi-LSTM-BERT [5]	0.82678	0.81945	0.83442	0.85792	0.81038
XGboost + Lexicon [6]	0.82156	0.81432	0.82324	0.86823	0.78145
PCA + K-means + Snow [7]	0.80897	0.80456	0.80986	0.83315	0.78864
DC-BiLSTM-CNN [8]	0.80673	0.79987	0.80679	0.82134	0.79256
Ekman-BERT [9[	0.84257	0.83412	0.83995	0.87412	0.80987
Our method	**0.87533**	**0.89870**	**0.86966**	**0.92426**	**0.83049**

Table 19
Evaluation index values corresponding to comparison methods based on search product online review data.

Method	AUC	ACC	F1	Precision	Recall
RoBERTa [1]	0.87523	0.85637	0.86664	0.89214	0.84386
ALiteBERT [2]	0.85346	0.83274	0.84464	0.87165	0.82037
XLNET [3]	0.86785	0.84736	0.85768	0.88453	0.83516
ELECTRA + CNN [4]	0.83912	0.82159	0.83231	0.86324	0.80348
SA-attention-Bi-LSTM-BERT [5]	0.89638	0.87546	0.88205	0.91768	0.85135
XGboost-Lexicon [6]	0.86027	0.83982	0.85757	0.89987	0.81975
PCA + K-means + Snow [7]	0.82954	0.80987	0.81463	0.84012	0.78965
DC-BiLSTM-CNN [8]	0.88457	0.86982	0.88012	0.91478	0.84963
Ekman-BERT [9]	0.92786	0.91987	0.91585	0.95532	0.87645
Our method	**0.94413**	**0.95670**	**0.92381**	**0.97044**	**0.88748**

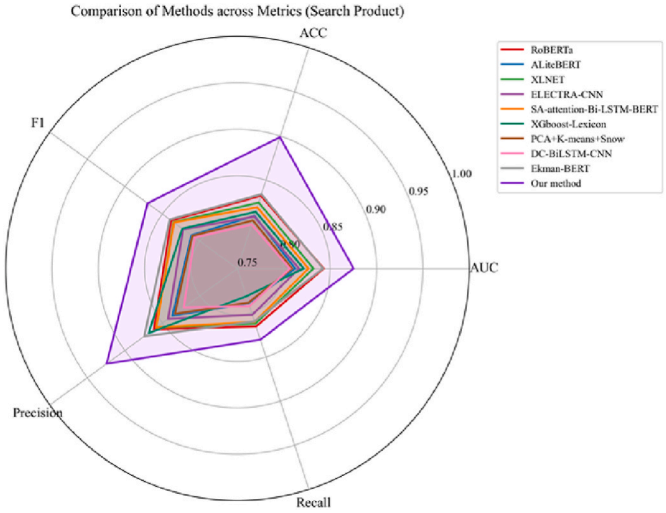


Fig. 8. The comparison results of the corresponding evaluation index for all comparison methods based on the experience product online review data.

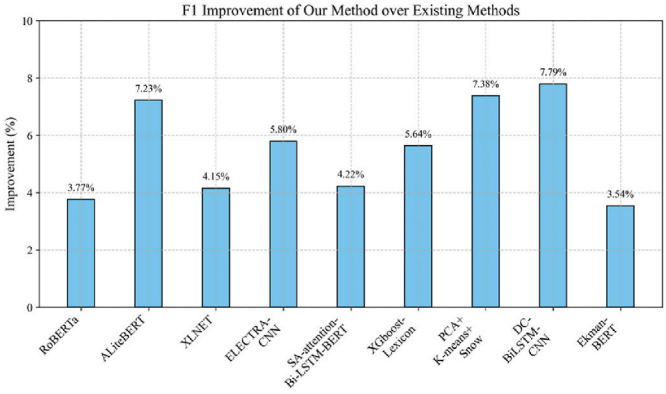


Fig. 9. The improvement rate of the evaluation index F1-Score of our method relative to each comparison method based on the experience product online review data.

4.5.3. Results analysis on the search product data

The results in Table 19 further confirm the cross-domain adaptability

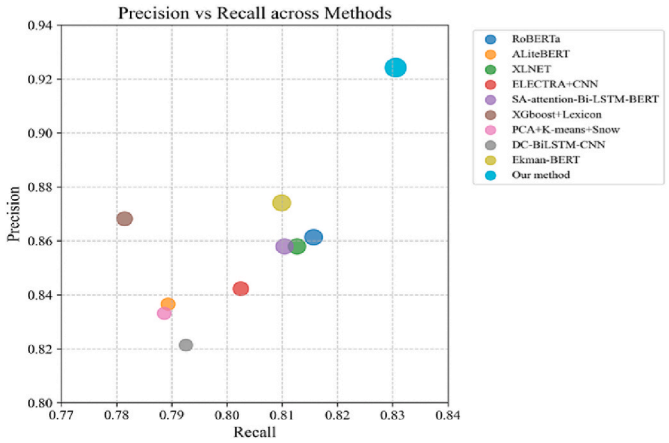


Fig. 10. The Precision, Recall, and F1 tradeoff renderings of all comparison methods based on the experience product online review data.

of our method. On the search-oriented product dataset, our method demonstrates exceptional performance across all metrics: achieving an AUC of 0.94413, an ACC of 0.95670, an F1 of 0.92381, a Precision of 0.97044, and a Recall of 0.88748. Each metric exceeds the performance of the best existing methods, demonstrating the robustness and effectiveness of our approach. To further illustrate these performance differences, we visualized the results from Table 19 in Fig. 11 through 13.

Multidimensional comparison in Fig. 11 highlights the superior performance of our method on the search-oriented product dataset, with the radar chart showing a significantly larger covered area compared to other methods. Fig. 12 demonstrates the universality of the F1-score improvement: even when compared to the best-performing Ekman-BERT, our method achieves a 0.87 % improvement, while the improvement reaches 13.40 % relative to the weakest-performing PCA + K-means + Snow. Fig. 13 further confirms, through a bubble chart, the notable ability of our method to balance Precision, Recall, and F1.

The comprehensive experimental results indicate that the proposed method achieves significant performance improvements under the multidimensional evaluation framework. This advantage has been validated on two datasets with distinct characteristics, demonstrating the robustness and generalizability of the method.

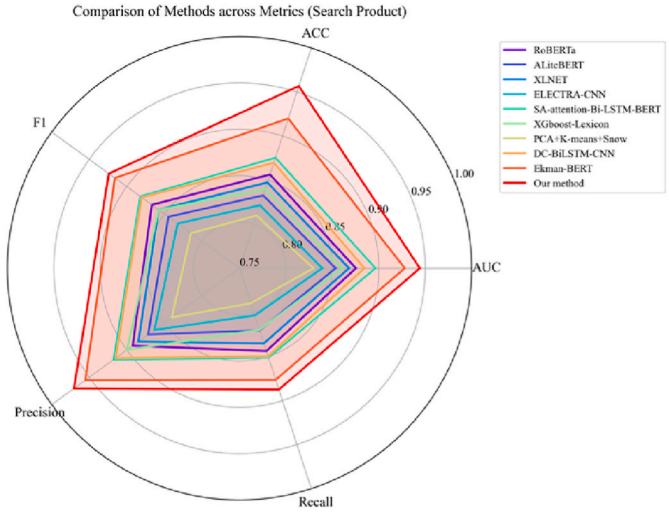


Fig. 11. The comparison results of the conrespondind evalution index for all comparison methods based on the search product online review data.

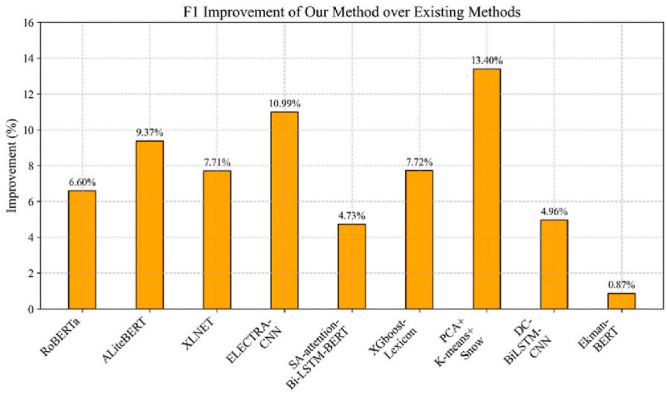


Fig. 12. The improvement rate of the evaluation index F1 of this method relative to each comparison method based on the search product online review data.

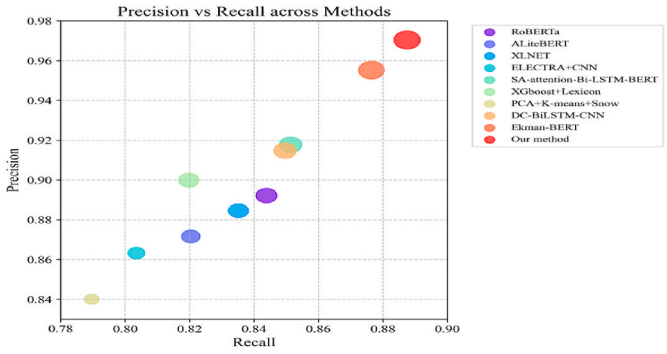


Fig. 13. The Precision, Recall, and F1 tradeoff renderings of all comparison methods based on the experience product online review data.

4.6. Relability analysis of the final weight and rank for the features

For the above results, this section will verify the validity and reliability of the final results by means of comparison. Firstly, the weights obtained based on traditional methods and those obtained based on perceived useful information are respectively brought into TOPSIS to obtain the ranking of each feature, and the results are compared and analyzed to further verify the effectiveness of our method.

First, we compared the BWM method with the entropy weight method (EWM) to verify that choosing the BWM method to calculate weights is scientifically effective. Therefore, we used the BWM and EWM methods to calculate the weights of features in traditional methods and the features in our method, as shown in Fig. 14. According to the results shown in the figure, by changing the data input into EWM, the results will change significantly, while the results will change very little by changing the data input into BWM. This means that the stability of calculating feature weights using the BWM method is higher than that of EWM, so using the BWM method to calculate weights is more effective.

Further, we introduce the consistency ratio, which verifies that the weights obtained based on perceived useful information better reflect the importance of features by comparing the weights based on the original data with those based on perceived useful information. The formula for calculating the consistency ratio is as follows:

$$R^C = \xi^* / I^C, \tag{1}$$

where, R^C denotes the consistency ratio, ξ^* is the optimal value of the objective function ξ, I^C refers to the consistency index, its calculation formula is as follows:

$$I^C = 0.5 + a_{bw}. \tag{2}$$

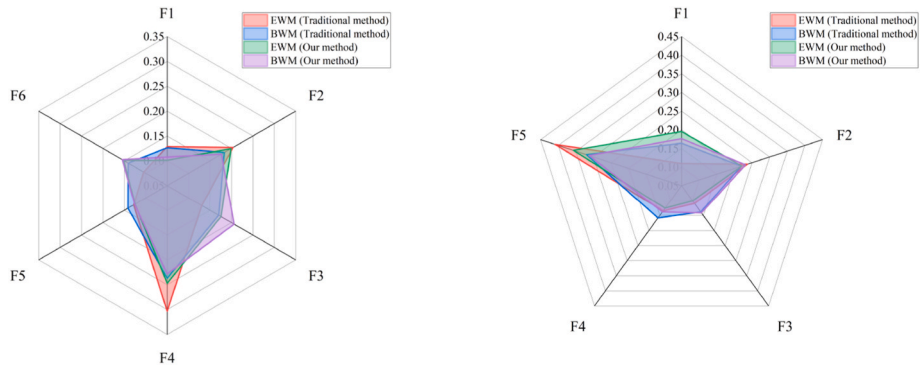


Fig. 14. Feature weights, obtained using BWM and EWM, where the left figure shows the weights for the first to sixth features of the experience product (F1 to F6 in Table 17), and the right figure shows the weights for the first to fifth features of the search product (F1 to F5 in Table 17).

Here, a_{bw} is an integer that is not less than 1. Obviously, before computing R^C, the value of I^C should first be calculated based on the a_{bw} determined by the subscripts b and w of the optimal index and the worst index. Therefore, based on the statistical data of products in Tables 11 and 12, we can obtain the optimal and worst criteria of sub-product features and product features under traditional methods and our method, as shown in Fig. 15.

Based on the results in Fig. 15, by calculating the ratio of the optimal criteria and other criteria in the corresponding statistical data in Tables 11 and 12, we can obtain the two-by-two comparison values shown in Fig. 16. Then, combining the results in these two figures, we can determine the values of a_{bw}, I^C, and ξ^*. The results are recorded in Fig. 17. Based on the results in Fig. 17, it is taken into equation (1) to obtain the consistency ratio in Fig. 18.

According to the results shown in Fig. 18, the reliability level of the final output of the traditional method is lower than that of the final output of our method. This means that our method has a higher reliability level of the final output based on the features and their weights calculated based on perceived useful information, which better meets the consumer market demand.

Finally, the weights obtained from the traditional method and the weights obtained from perceived useful information were respectively input into TOPSIS to obtain the ranking of each feature, as shown in Fig. 19. According to the results shown in the figure, for the experiential products, the feature ranking obtained by inputting the weights obtained from the traditional method into TOPSIS has some deviation from that obtained by inputting the weights obtained from perceived useful information into TOPSIS, but the optimal feature is F5, the suboptimal features are all F4, and the worst feature is the same, which is F3. For the search products, the feature ranking obtained by inputting the weights obtained from the traditional method into TOPSIS is the same as that obtained by inputting the weights obtained from perceived useful information into TOPSIS. However, it is worth noting that the key attributes obtained by combining the traditional method or our method with TOPSIS are the same for both types of products. Therefore, the optimal decision obtained by combining the weights obtained from perceived useful information method with TOPSIS is reliable and effective.

5. Discussions

The new CPUI prediction method proposed in this study, which combines information entropy with machine learning, provides a practical technical pathway for addressing the problem of information overload in online review analysis. By extracting only effective information, this approach can quickly identify key reviews and supply more precise data support for subsequent feature extraction and preference weight computation. Taking the real data case study results in this article as an example, only 45.291 % of reviews for experience products (toothpaste) and 56.722 % for search products (medical apps) were recognized as truly perceived useful information by consumers. In contrast, traditional methods typically analyze all review data, leading to a mix of a large amount of redundant and potentially misleading information, which in turn affects the overall accuracy and efficiency of the analysis. The method proposed in this study outperforms baseline methods such as voting, significantly improving data processing efficiency and reducing the interference of noise. This provides decision-makers with a reliable basis that accurately reflects consumers' true needs, ensuring that companies operating in resource-constrained and data-intensive markets can rapidly capture core insights and precisely guide product design and operational strategies.

The CPUI-based feature extraction method proposed in this study not only significantly improves the capture of dynamic changes in consumer preferences but also identifies hidden product features that traditional methods struggle to extract. For example, in the analysis of experience products (toothpaste) using our data, conventional methods mainly focus on surface-level attributes, such as "packaging box" or "ease of opening," whereas our method—after filtering with high information entropy—can uncover more detailed features such as "Repair" and "The

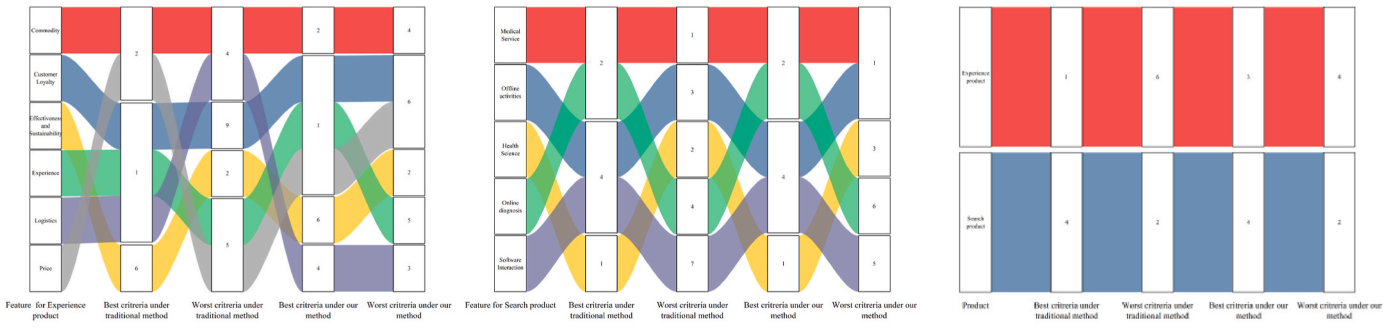


Fig. 15. Optimal and worst-case criteria for sub-product features and product features under traditional methods and our method.

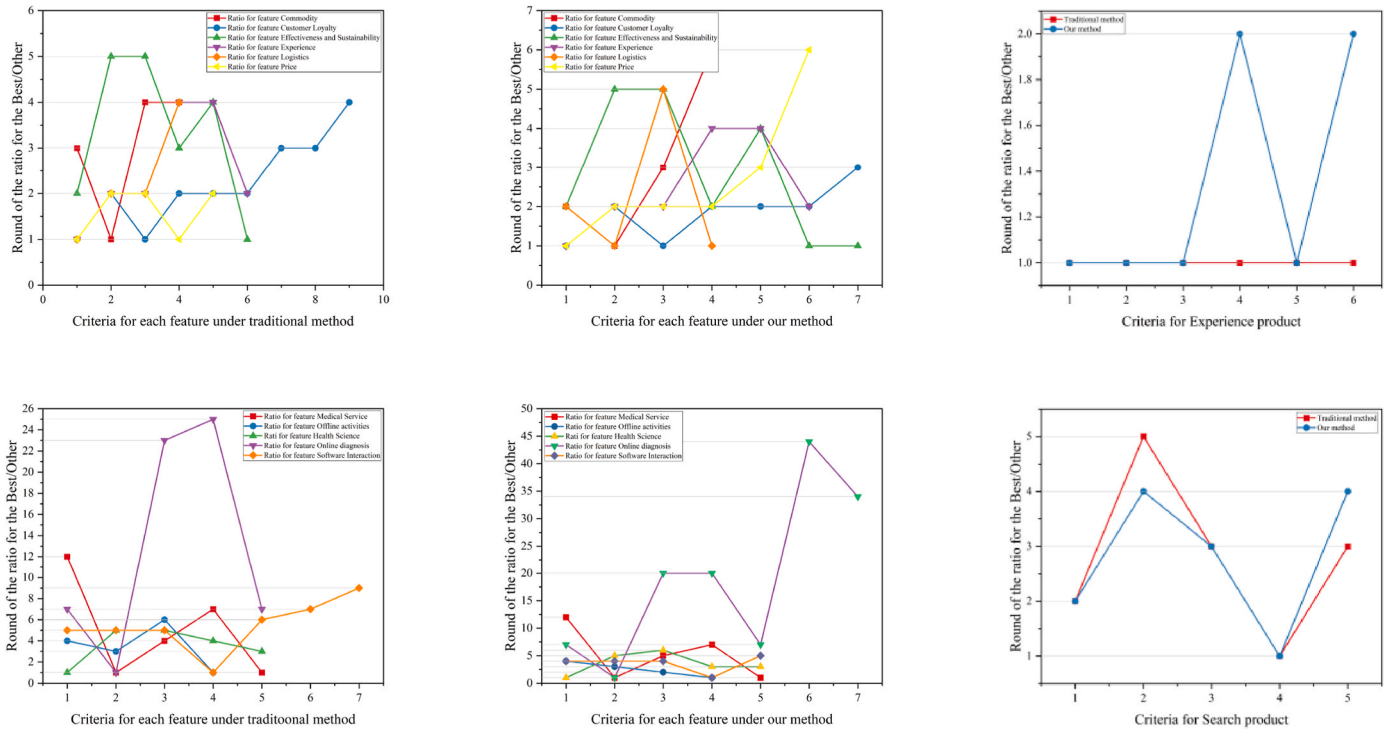


Fig. 16. Pairwise comparison value for the best criterion, where the top three figures correspond to Experience product, and the bottom three figures correspond to Search product.

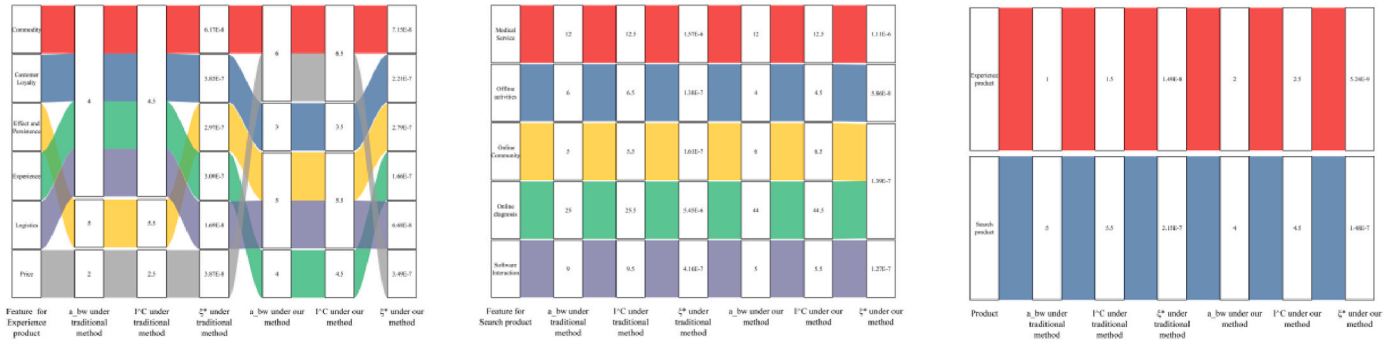


Fig. 17. The values of a_{bw}, I^C and ξ^* for sub-product features and product features under the conventional method and our method.

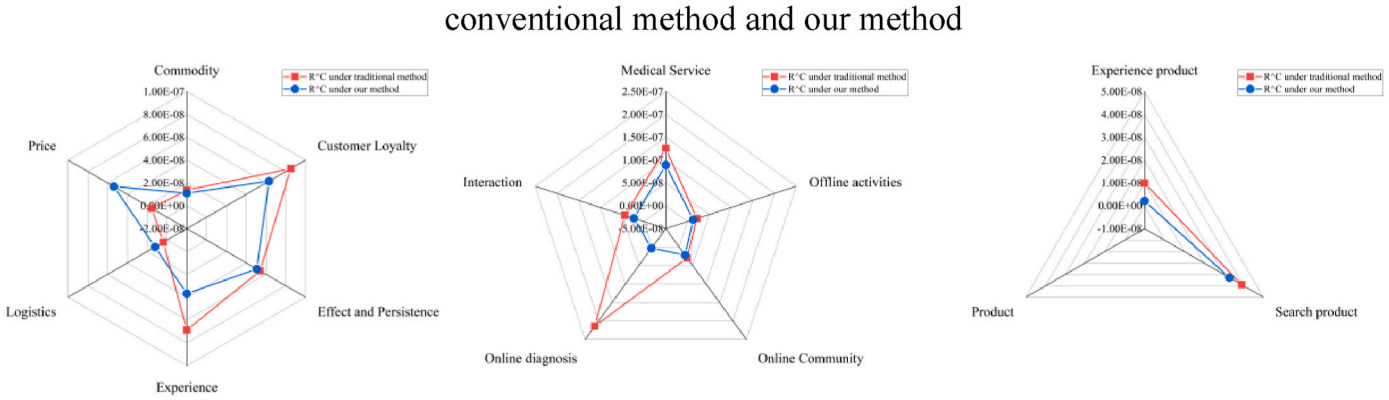


Fig. 18. The consistency ratio of sub-product features and product features under traditional methods and our method, where the consistency ratio of experience-based products, search-based products, and the overall product is shown from left to right.

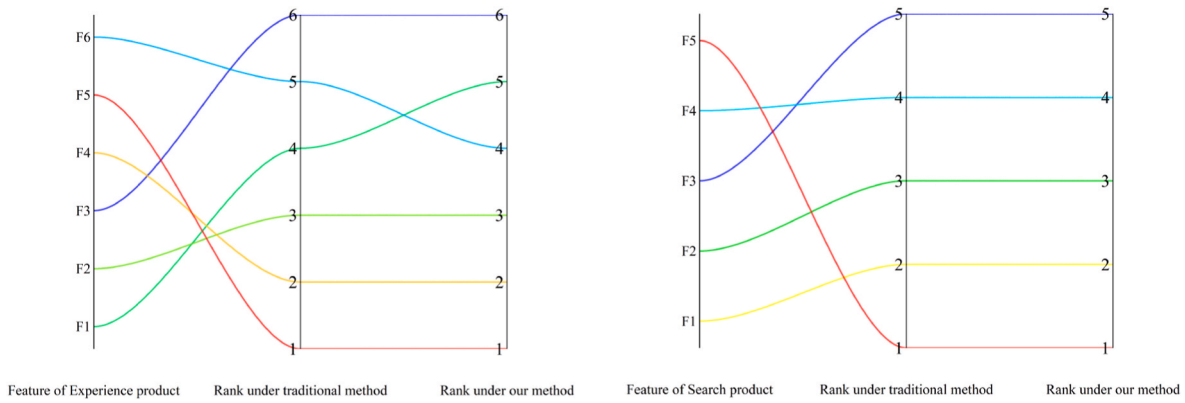


Fig. 19. The ranking of features for different types of products in the traditional method and our method.

second day," revealing that consumers place greater emphasis on the product's sustained efficacy and after-effects. In the case of search products (medical apps), our approach successfully extracts consumer-focused features related to "illness" descriptions, clearly indicating that users are increasingly sensitive to new demands in product diagnosis and risk notification. This ensures that companies can gain a more comprehensive insight into changing market demands and, based on these implicit features, adjust their product development and marketing strategies in a timely manner. Ultimately, this provides robust data support for product innovation and brand upgrading, further enhancing the company's long-term competitive advantage.

The CPUI-based weight correction method proposed in this study not only more accurately reflects the true attention consumers pay to various product features, but also provides clear, actionable decision support for companies in their day-to-day operations. This ensures that firms can accurately identify the key value of each product feature in practice, and subsequently focus on optimizing the core features that truly impact consumer satisfaction and loyalty in product development, supply chain management, and marketing strategy adjustments. For example, based on the actual data in this study, in toothpaste (an experience product) the weight for the "product efficacy" feature increases significantly from 0.170 (traditional method) to 0.206(our method), with its rank improving from third to second; meanwhile, the weight for the "logistics and delivery" feature decreases from 0.142 to 0.126, with its rank dropping from fourth to fifth. This indicates that traditional methods tend to overestimate peripheral factors (such as delivery speed) while underestimating the features that truly affect consumer satisfaction (such as sustained efficacy). In the case of medical apps (search products), our method shows that "software interaction" remains the highest weighted feature (0.312), whereas the weight for "online diagnosis" decreases from 0.158 to 0.138, with its ranking correspondingly lowered. This clearly reflects consumers' higher emphasis on the interactive experience, thereby providing a basis for more accurately capturing true consumer needs. For instance, for toothpaste products, companies can adjust their R&D and production resources based on the revised weights—allocating relatively more resources toward enhancing product efficacy and durability—whereas, for medical apps, firms can concentrate on optimizing software interaction and diagnostic functionality, effectively reducing potential risks and boosting user trust.

In summary, our method offers managers in dynamic markets a more efficient, precise, and targeted decision-support that not only enables comprehensive strategic resource optimization in a big data environment, but also dynamically and accurately enhances consumer satisfaction and brand competitiveness to establish a sustainable competitive advantage.

5.1. Theoretical implications

First, our proposed CPUI prediction model combines Shannon's information entropy with a machine learning binary classification approach. This method addresses the inherent biases in traditional voting methods and establishes a more robust approach to evaluate the informational value of online reviews, thereby refining the theoretical framework for quantifying the quality of consumer feedback.

Second, our comprehensive hidden feature extraction method integrates both dynamic and implicit feedback to uncover emerging product attributes and latent consumer demands that are often overlooked by conventional methods. This work enriches the literature on how explicit and implicit features influence consumer decision-making and advances the development of consumer preference theory.

Finally, our novel weight calculation method combines CPUI-based filtering with an optimized BWM-TOPSIS framework. By reducing the impact of redundant and invalid data, this method precisely associates consumer feedback with feature prioritization in decision-making models, providing a novel theoretical perspective for weight analysis.

In summary, these contributions address key gaps in current research and strengthen the theoretical foundation for extracting consumer preferences in the era of big data.

5.2. Managerial implications

The new CPUI prediction model enhances the identification of valuable consumer feedback by accurately distinguishing useful reviews from noise. In practice, this empowers managers to rely on high-quality data to tailor product designs and marketing strategies. By focusing on reviews that truly reflect consumer sentiment, companies can reduce misallocation of resources and make more informed decisions in real time.

The comprehensive hidden feature extraction method uncovers emerging product attributes and latent consumer demands that conventional approaches often miss. This capability allows businesses to detect subtle shifts in consumer preferences early—enabling agile adjustments in product innovation and strategy. In effect, it paves the way for companies to proactively meet market demands and stay ahead of competitors.

The novel weight calculation method offers a more objective means of prioritizing product features based on consumer-perceived usefulness. By filtering out redundant or invalid data, this approach provides more accurate weightings that reflect true consumer preferences. Consequently, managers can optimize resource allocation by investing in product features that are most likely to enhance customer satisfaction and strengthen brand loyalty.

Collectively, these contributions not only advance theoretical understanding but also furnish managers with a robust, practical decision-

support tool. This tool aids in refining product design, adapting marketing strategies, and improving resource allocation based on reliable insights from consumer data.

5.3. Limitations and future research directions

Although our results outperform the current SOTA algorithms, this study did not conduct an in-depth investigation into cross-cultural and regional differences. In future work, online review data from different countries and regions can be analyzed to examine variations in consumer attention to product features, thereby providing more targeted product design and marketing strategy recommendations for global enterprises.

Furthermore, although our current study focuses on two product categories—experience products (e.g., toothpaste reviews) and search products (e.g., medical app reviews)—it may not fully capture the broader spectrum of online review behavior. Future research could incorporate more product types, which would help to more comprehensively evaluate the applicability of our framework across different product categories, enhance the generality of the results, and offer broader applications for sentiment impact analysis.

Finally, while text sentiment analysis plays a crucial role in the development of interactive AI, the scope of this study limits an in-depth exploration of its broader applications. Future research could explore how the proposed methodology can be integrated into AI system development to enable high-quality information interactions that are more in line with user preference needs, thereby increasing the accuracy of AI-driven interactions and improving the decision-making process.

CRediT authorship contribution statement

Zheng Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Huiran Liu:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis. **Xiaojun Fan:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that there is no conflict of interest in their work.

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Data availability

Data will be made available on request.

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